

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402443
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Moon; Su Mi

Light-emitting element, method for manufacturing same, and display device

Abstract

A light-emitting element, a method for manufacturing same, and a display device are provided. The light-emitting element comprises: a first semiconductor layer and a second semiconductor layer; an active layer between the first semiconductor layer and the second semiconductor layer; and an electrode layer on the second semiconductor layer and having a first surface and a second surface facing the first surface, the second surface contacting the second semiconductor layer. The electrode layer includes a first protruding pattern in which at least a partial area of the first surface protrudes.

Inventors:	Moon; Su Mi (Suwon-si, KR)
Applicant:	Samsung Display Co., LTD. (Yongin-si, KR)
Family ID:	1000008782895
Assignee:	Samsung Display Co., Ltd. (Yongin-si, KR)
Appl. No.:	17/575453
Filed (or PCT Filed):	February 27, 2020
PCT No.:	PCT/KR2020/002801
PCT Pub. No.:	WO2021/010563
PCT Pub. Date:	January 21, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230282777 A1	Sep. 07, 2023

Foreign Application Priority Data

Publication Classification

Int. Cl.: H10H20/83 (20250101); H10H20/82 (20250101); H10H20/831 (20250101);
H10H20/857 (20250101)

U.S. Cl.:

CPC H10H20/8312 (20250101); H10H20/82 (20250101); H10H20/857 (20250101);

Field of Classification Search

CPC: H01L (33/382); H01L (33/22); H01L (33/62); H01L (33/0095); H01L (25/0753); H01L (33/20); H01L (33/38); H01L (21/28); H01L (25/167); H10H (20/8312); H10H (20/82); H10H (20/857); H10H (20/831); H10H (20/819); H10H (20/01); H10H (20/032); H10H (29/142); H10D (8/043)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8115224	12/2011	Kim	N/A	N/A
10672946	12/2019	Cho et al.	N/A	N/A
11114582	12/2020	Cho et al.	N/A	N/A
12040425	12/2023	Cho et al.	N/A	N/A
2010/0181586	12/2009	Kim	257/E33.064	H10H 20/841
2011/0124139	12/2010	Chang	438/40	H10H 20/018
2013/0119423	12/2012	Lee	257/98	H10H 20/855
2015/0069444	12/2014	Chae	257/98	H10H 20/84
2015/0076535	12/2014	Choung	N/A	N/A
2018/0023793	12/2017	Lowenthal	362/296.09	H10H 20/855
2022/0149241	12/2021	Lee	N/A	H10H 20/8312

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101783382	12/2009	CN	N/A
204243076	12/2014	CN	N/A
107611153	12/2017	CN	N/A
10-2010-0085578	12/2009	KR	N/A
10-2011-0115384	12/2010	KR	N/A
20110115384	12/2010	KR	N/A

10-2013-0052825	12/2012	KR	N/A
10-2015-0035211	12/2014	KR	N/A
20150035211	12/2014	KR	N/A
10-2016-0039172	12/2015	KR	N/A
10-2016-0059574	12/2015	KR	N/A
10-1742518	12/2016	KR	N/A
1323520	12/2009	TW	N/A
WO 2012/148231	12/2011	WO	N/A
WO 2018/117073	12/2017	WO	N/A

OTHER PUBLICATIONS

Chinese Office Action for CN Application No. 202080050853.X, dated Aug. 23, 2024, 13 pages. cited by applicant

Extended European Search Report issued in corresponding Application No. EP 20841155.3, dated Jul. 14, 2023, 14 pages. cited by applicant

International Search Report for corresponding Application No. PCT/KR2020/002801 dated Jun. 17, 2020, 4pp. cited by applicant

Primary Examiner: Sohel Imtiaz; S M

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a U.S. National Phase Patent Application of International Patent Application Number PCT/KR2020/002801, filed on Feb. 27, 2020, which claims priority to Korean Patent Application Number 10-2019-0086383, filed on Jul. 17, 2019, the entire content of all of which is incorporated herein by reference.

BACKGROUND

1. Field

(2) Aspects of embodiments of the present disclosure relate to a light-emitting element, a method for manufacturing the same, and a display device.

2. Description of the Related Art

(3) Display devices are increasingly important with the development of multimedia. Accordingly, various types of display devices such as organic light-emitting diode (OLED) display devices and liquid crystal display (LCD) devices are being used.

(4) A display panel, such as an OLED panel or an LCD panel, is a device included in a display device to (e.g., that is configured to) display an image. From among such display panels, a light-emitting diode (LED) may be provided as a light-emitting display element, and examples of a light-emitting diode (LED) include an organic LED (OLED) using an organic material as a fluorescent material and an inorganic LED using an inorganic material as a fluorescent material.

(5) An inorganic LED using an inorganic semiconductor as a fluorescent material has durability even in a high-temperature environment and more efficiently emits blue light compared to the organic LED. A transfer method using a dielectrophoresis (DEP) method has been developed as part of a manufacturing process for existing inorganic LED elements, but this process may be limiting. Accordingly, research is being continuously conducted on the inorganic light-emitting diode having higher durability and efficiency than those of the organic light-emitting diode.

SUMMARY

- (6) Aspects of embodiments of the present disclosure provide a light-emitting element in which both end surfaces (e.g., opposite end surfaces) have a different roughness and a pattern is formed on an end surface and a method for manufacturing the light-emitting element.
- (7) Aspects of embodiments of the present disclosure provide a display device which includes a plurality of light-emitting elements, each having a different pattern formed on an end surface.
- (8) It should be noted that aspects and features of the present disclosure are not limited thereto and other aspects and features not mentioned herein will be apparent to those of ordinary skill in the art from the following description.
- (9) According to an embodiment of the present disclosure, a light-emitting element includes: a first semiconductor layer and a second semiconductor layer; an active layer between the first semiconductor layer and the second semiconductor layer; and an electrode layer on the second semiconductor layer and having a first surface and a second surface facing the first surface, the second surface contacting the second semiconductor layer. At least any one of the first surface and the second surface of the electrode layer includes a protrusion pattern protruding from at least a portion thereof.
- (10) The protrusion pattern may include: a first protrusion, a second protrusion spaced apart from the first protrusion, and a depression between the first protrusion and the second protrusion.
- (11) The second protrusion may extend around a periphery of the first protrusion.
- (12) The protrusion pattern may further include a third protrusion spaced apart from the first protrusion and the second protrusion.
- (13) The third protrusion may extend around a periphery of an outer surface of the second protrusion, and the depression may be between the second protrusion and the third protrusion.
- (14) The first protrusion may have an area having the same width as that of the second protrusion.
- (15) The first protrusion may have an area having a width equal to a distance between the first protrusion and the second protrusion.
- (16) The protrusion pattern may include a first protrusion pattern formed on the first surface, and the light-emitting element may extend in a direction and has a first end surface on which the first protrusion pattern is formed and a second end surface which is a surface of the first semiconductor layer.
- (17) The first end surface and the second end surface may have different roughness.
- (18) The light-emitting element may further include an insulating film which at least partially surrounds side surfaces of the first semiconductor layer, the second semiconductor layer, and the active layer.
- (19) The electrode layer may include a second protrusion pattern which is formed on the second surface to contact the second semiconductor layer, and at least a portion of a surface of the second semiconductor layer that contacts the second surface of the electrode layer may be depressed to contact protrusions of the second protrusion pattern.
- (20) The protrusion pattern may also be formed on the first surface.
- (21) According to an embodiment of the present disclosure, a method for manufacturing light-emitting elements includes: providing a semiconductor structure on a substrate; and forming a plurality of element rods, each of the element rods including a protrusion pattern protruding from at least a portion of an upper surface, by etching the semiconductor structure. Different protrusion patterns are formed in different areas of the semiconductor structure.
- (22) The forming of the element rods may include: forming a first pattern layer including first mask patterns spaced apart from each other on the semiconductor structure; forming semiconductor crystals by etching the semiconductor structure along an area between the first mask patterns spaced apart from each other; forming a second pattern layer including second mask patterns spaced apart from each other on the semiconductor crystals; and forming element rods including protrusion patterns by partially etching upper surfaces of the semiconductor crystals along an area

in which the second mask patterns are spaced apart from each other.

(23) The forming of the second pattern layer may be performed after an insulating film surrounding an outer circumferential surface of each of the semiconductor crystals is formed.

(24) The semiconductor structure may have a first area and a second area different from the first area, and the element rods may include a first element rod formed in the first area and a second element rod formed in the second area.

(25) In the first pattern layer, the first mask pattern in the first area and the first mask pattern in the second area may have the same shape.

(26) The first element rod and the second element rod may have the same diameter.

(27) In the second pattern layer, the second mask pattern in the first area and the second mask pattern in the second area may have different shapes.

(28) A first protrusion pattern formed on an upper surface of the first element rod may have a shape different from that of a second protrusion pattern formed on an upper surface of the second element rod.

(29) The forming of the element rods may include: forming a third mask layer including third mask patterns spaced apart from each other on the semiconductor structure; forming protrusion patterns by partially etching an upper surface of the semiconductor structure along an area in which the third mask patterns are spaced apart from each other; forming a fourth mask layer including fourth mask patterns spaced apart from each other on the semiconductor structure; and forming element rods by etching the semiconductor structure along an area between the fourth mask patterns spaced apart from each other.

(30) The first area and the second area may have different light emission amounts.

(31) According to an embodiment of the present disclosure, a display device includes: a first electrode and a second electrode spaced apart from the first electrode; and a light-emitting element between the first electrode and the second electrode and comprising a pattern part formed on an end surface. The light-emitting element includes: a first semiconductor layer and a second semiconductor layer; an active layer between the first semiconductor layer and the second semiconductor layer; and an electrode layer on the second semiconductor layer and including a protrusion pattern protruding from at least a portion of a surface.

(32) The protrusion pattern may include a first protrusion, a second protrusion spaced apart from the first protrusion, and a depression between the first protrusion and the second protrusion.

(33) The second protrusion may extend around a periphery of the first protrusion.

(34) The light-emitting element may include a first light-emitting element including a first protrusion pattern and a second light-emitting element including a second protrusion pattern different from the first protrusion pattern.

(35) The first protrusion pattern may have a different number of protrusions than the second protrusion pattern.

(36) A width of each protrusion in the first protrusion pattern may be different from a width of each protrusion in the second protrusion pattern.

(37) A first end of the light-emitting element may be electrically connected to the first electrode, and a second end may be electrically connected to the second electrode.

(38) The electrode layer of the light-emitting element may be electrically connected to the first electrode, and the first semiconductor layer may be electrically connected to the second electrode.

(39) The display device may further include: a first contact electrode contacting the first end of the light-emitting element and the first electrode; and a second contact electrode contacting the second end of the light-emitting element and the second electrode.

(40) The first contact electrode may contact the electrode layer of the light-emitting element, and the second contact electrode may contact the first semiconductor layer.

(41) Details of other embodiments of the present disclosure are included in the detailed description and the accompanying drawings.

(42) In a method for manufacturing light-emitting elements according to an embodiment, light-emitting elements having protrusion patterns may be manufactured through a plurality of (e.g., two) etching processes for etching a semiconductor structure. The light-emitting elements may include (or have) different protrusion patterns, and positions at where the light-emitting elements are formed in the semiconductor structure may be identified through (e.g., by using) the protrusion patterns. Accordingly, when there is a difference between light-emitting characteristics of the manufactured light-emitting elements, the positions at where the light-emitting elements are formed in the semiconductor structure can be identified through the protrusion patterns, and light-emitting characteristics at the positions can be modified and/or controlled.

(43) A display device according to an embodiment may include light-emitting elements having different protrusion patterns to reduce a difference in light-emitting characteristics to have uniform (or substantially uniform) light-emitting characteristics in each pixel.

(44) The above-described aspects and features do not limit those of the present disclosure, and other various aspects and feature are included in this disclosure or will be apparent to those of ordinary skill in the art from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic plan view of a display device according to an embodiment;
- (2) FIG. 2 is a schematic plan view of a pixel of the display device shown in FIG. 1;
- (3) FIG. 3 is a plan view of a subpixel shown in FIG. 2;
- (4) FIG. 4 is a cross-sectional view taken along the lines Xa-Xa', Xb-Xb' and Xc-Xc' of FIG. 3;
- (5) FIG. 5 is a cross-sectional view of a part of a display device according to an embodiment;
- (6) FIG. 6 is a schematic view of a light-emitting element according to an embodiment;
- (7) FIG. 7 is a cross-sectional view taken along the line VI-VI' of FIG. 6;
- (8) FIG. 8 is a cross-sectional view taken along the line VII-VII' of FIG. 3;
- (9) FIG. 9 is a flowchart illustrating a method for manufacturing light-emitting elements according to an embodiment;
- (10) FIGS. 10 through 25 are schematic views illustrating a process of manufacturing light-emitting elements according to an embodiment;
- (11) FIGS. 26 through 28 are plan views illustrating shapes of a second pattern layer according to embodiments;
- (12) FIGS. 29 and 30 are cross-sectional views illustrating steps of a process of manufacturing light-emitting elements according to an embodiment;
- (13) FIGS. 31a and 31b are cross-sectional views of a light-emitting element according to an embodiment;
- (14) FIG. 32 is a schematic view of a light-emitting element according to an embodiment;
- (15) FIG. 33 is a cross-sectional view taken along the line VI-VI' of FIG. 32;
- (16) FIG. 34 is a plan view of a subpixel of a display device according to an embodiment; and
- (17) FIG. 35 is a plan view of a pixel of a display device according to an embodiment.

DETAILED DESCRIPTION

(18) Aspects and features of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the present disclosure are shown. The present disclosure may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art.

(19) It will be understood that when a layer is referred to as being “on” another layer (or substrate),

it may be directly on the other layer or substrate or one or more intervening layers may be present. Similarly, it will also be understood that when an element is referred to as being “on,” “connected to,” or “coupled to” another element, it may be directly on, connected, or coupled to the other element or one or more intervening elements may be present. When an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. For example, when a first element is described as being “coupled” or “connected” to a second element, the first element may be directly coupled or connected to the second element or the first element may be indirectly coupled or connected to the second element via one or more intervening elements.

(20) In the figures, dimensions of the various elements, layers, etc. may be exaggerated for clarity of illustration. The same reference numerals designate the same elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the use of “may” when describing embodiments of the present disclosure relates to “one or more embodiments of the present disclosure.” As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. As used herein, the terms “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

(21) It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. For instance, a first element discussed below could be termed a second element without departing from the teachings of the invention. Similarly, the second element could also be termed the first element.

(22) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” or “over” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein should be interpreted accordingly.

(23) The terminology used herein is for the purpose of describing particular example embodiments of the present disclosure and is not intended to be limiting of the described example embodiments of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(24) Hereinafter, embodiments will be described with reference to the accompanying drawings.

(25) FIG. 1 is a schematic plan view of a display device according to an embodiment.

(26) Referring to FIG. 1, a display device **10** displays (e.g., is configured to display) moving images or still images. The display device **10** may refer to any electronic device that provides a display screen. Examples of the display device **10** may include televisions, notebook computers, monitors, billboards, an Internet of things (IoT) device, mobile phones, smartphones, tablet personal computers (PCs), electronic watches, smart watches, watch phones, head mounted displays, mobile communication terminals, electronic notebooks, electronic books, portable multimedia players (PMPs), navigation devices, game machines, digital cameras, camcorders, and

other devices that provide (or include) a display screen.

(27) The display device **10** includes a display panel that provides a display screen. Examples of the display panel may include a light emitting diode display panel, an organic light emitting diode display panel, a quantum dot light emitting display panel, a plasma display panel, and a field emission display panel. An embodiment in which a light emitting diode display panel is applied as the display panel will be described below, but the present disclosure is not limited to this embodiment. Other display panels may be applied as the display panel as long as the same technical spirit is applicable.

(28) The shape of the display device **10** can be variously modified. For example, the display device **10** may have various shapes, such as a horizontally long rectangle (e.g., a horizontally extended rectangle), a vertically long rectangle (e.g., a vertically extended rectangle), a square, a quadrangle with rounded corners (or vertices), other polygons, and a circle. The shape of a display area DA of the display device **10** may also be similar to the overall shape of the display device **10**. In FIG. **1**, the display device **10** and the display area DA are illustrated as having a horizontally long rectangular shape.

(29) The display device **10** may have the display area DA and a non-display area NDA. The display area DA may be an area where a screen is present (e.g., where an image can be displayed), and the non-display area NDA may be an area where no image is displayed. The display area DA may also be referred to as an active area, and the non-display area NDA may also be referred to as an inactive area.

(30) The display area DA may generally occupy the center of the display device **10**. The display area DA may include a plurality of pixels PX. The pixels PX may be arranged in a matrix pattern. Each of the pixels PX may have a rectangular or square shape in a plan view. However, the present disclosure is not limited thereto, and each of the pixels PX may also have a rhombus shape in which each side is inclined with respect to a direction. Each of the pixels PX may display (or emit) a specific color by including one or more light-emitting elements **300** which emit light of (or in) a specific wavelength band.

(31) FIG. **2** is a schematic plan view of a pixel of the display device according to the embodiment shown in FIG. **1**, and FIG. **3** is a plan view of a subpixel shown in FIG. **2**.

(32) Referring to FIGS. **2** and **3**, each of the pixels PX may include a first subpixel PX1, a second subpixel PX2, and a third subpixel PX3. The first subpixel PX1 may emit light of (or having) a first color, the second subpixel PX2 may emit light of a second color, and the third subpixel PX3 may emit light of a third color. The first color may be blue, the second color may be green, and the third color may be red. However, the present disclosure is not limited thereto, and in other embodiments the subpixels PXn (e.g., all of the subpixels PXn) may emit light of the same color. Although a pixel PX is illustrated in FIG. **2** as including three subpixels PXn (i.e., first through third subpixels PX1, PX2, PX3), the present disclosure is not limited thereto, and the pixel PX may include more subpixels PXn.

(33) Each subpixel PXn of the display device **10** may have an area defined as an emission area EMA. The first subpixel PX1 may include a first emission area EMA1, the second subpixel PX2 may include a second emission area EMA2, and the third subpixel PX3 may include a third emission area EMA3. The emission area EMA may be defined as an area where light-emitting elements **300** are disposed to emit light of a specific wavelength band. Each of the light-emitting elements **300** may include an active layer **330** (see, e.g., FIG. **4**), and the active layer **330** may emit light of a specific wavelength band without directionality. For example, light emitted from the active layer **330** of each light-emitting element **300** may be radiated in a lateral direction of the light-emitting element **300** as well as toward both ends (e.g., opposite ends) of the light-emitting element **300**. The emission area EMA of each subpixel PXn may include an area in which the light-emitting elements **300** are disposed and an area which is adjacent to the light-emitting elements **300** and from which light emitted from the light-emitting elements **300** is output. In addition, the

present disclosure is not limited thereto, and the emission area EMA may also include an area from which light emitted from the light-emitting elements **300** is output after being reflected or refracted by other elements (e.g., other members). A plurality of light-emitting elements **300** may be disposed in each subpixel PXn, and an area where the light-emitting elements **300** are disposed and an area adjacent to this area may form the emission area EMA.

(34) Each subpixel PXn of the display device **10** may have a non-emission area defined as an area other than the emission area EMA. The non-emission area may be an area in which the light-emitting elements **300** are not disposed and from which no light is output because light emitted from the light-emitting elements **300** does not reach this area.

(35) Each subpixel PXn of the display device **10** may include a plurality of electrodes **210** and **220**, the light-emitting elements **300**, a plurality of contact electrodes **260**, a plurality of inner banks **410** and **420** (see, e.g., FIG. 4), an outer bank **430**, and one or more insulating layers **510**, **520**, and **550** (see, e.g., FIG. 4).

(36) The electrodes **210** and **220** may be electrically connected to the light-emitting elements **300** and may receive a voltage (e.g., a predetermined voltage) so that the light-emitting elements **300** emit light (e.g., emit light of a specific wavelength band). In addition, at least a part (or portion) of each of the electrodes **210** and **220** may be utilized to form an electric field in a subpixel PXn to align the light-emitting elements **300**.

(37) The electrodes **210** and **220** may include a first electrode **210** and a second electrode **220**. In an embodiment, the first electrode **210** may be a separate pixel electrode for each subpixel PXn, and the second electrode **220** may be a common electrode commonly connected along each subpixel PXn. Any one of the first electrode **210** and the second electrode **220** may be an anode of the light-emitting elements **300**, and the other may be a cathode of the light-emitting elements **300**. However, the present disclosure is not limited thereto, and the opposite may also be true.

(38) Each of the first electrode **210** and the second electrode **220** may have an electrode stem part **210S** or **220S** extending in a first direction DR1 and at least one electrode branch part **210B** or **220B** extending and branching from the electrode stem part **210S** or **220S** in a second direction DR2 crossing (e.g., intersecting) the first direction DR1.

(39) The first electrode **210** may include a first electrode stem part **210S** extending in the first direction DR1 and at least one first electrode branch part **210B** branching from the first electrode stem part **210S** and extending in the second direction DR2.

(40) The first electrode stem part **210S** of any one pixel may have both ends ending between subpixels PXn and spaced apart from ends of neighboring first electrode stem parts **210S** but may lie on substantially the same straight line as the first electrode stem parts **210S** of neighboring subpixels (e.g., adjacent in the first direction DR1) in the same row. Because both ends of the first electrode stem parts **210S** respectively disposed in the subpixels PXn are spaced apart from each other, a different electrical signal may be transmitted to each first electrode branch part **210B**, and each first electrode branch part **210B** may be driven separately.

(41) The first electrode branch part **210B** may branch from at least a part of the first electrode stem part **210S** and extend in the second direction DR2 to end at a position spaced apart from a second electrode stem part **220S** facing the first electrode stem part **210S**.

(42) The second electrode **220** may have the second electrode stem part **220S** extending in the first direction DR1 and spaced apart from the first electrode stem part **210S** in the second direction DR2 to face the first electrode stem part **210S** and a second electrode branch part **220B** branching from the second electrode stem part **220S** and extending in the second direction DR2. The other end of the second electrode stem part **220S** may be connected to the second electrode stem part **220S** of another subpixel PXn adjacent in the first direction DR1. For example, different from the first electrode stem part **210S**, the second electrode stem part **220S** may extend in the first direction DR1 to cross the subpixels PXn. The second electrode stem part **220S** crossing the subpixels PXn may be connected to a peripheral part of the display area DA in which each pixel PX or subpixel

PXn is disposed or a part extending in a direction in the non-display area NDA.

(43) The second electrode branch part **220B** may be spaced apart from the first electrode branch part **2106** to face the first electrode branch part **2106** and may end at a position spaced apart from the first electrode stem part **210S**. The second electrode branch part **220B** may be connected to the second electrode stem part **220S**, and an end in the extending direction may be spaced apart from the first electrode stem part **210S** in each subpixel PXn.

(44) The first electrode **210** and the second electrode **220** may be electrically connected to a circuit element layer PAL (see, e.g., FIG. 4) of the display device **10** through contact openings (e.g., contact holes), for example, a first electrode contact opening (e.g., a first electrode contact hole) CNTD and a second electrode contact opening (e.g., a second electrode contact hole) CNTS, respectively. In the drawings, the first electrode contact opening CNTD is formed in the first electrode stem part **210S** of each subpixel PXn, and only one second electrode contact opening CNTS is formed in one second electrode stem part **220S** crossing the subpixels PXn. However, the present disclosure is not limited thereto. In some embodiments, the second electrode contact opening CNTS may also be formed in each subpixel PXn.

(45) Although two first electrode branch parts **210B** are disposed in each subpixel PXn and one second electrode branch part **220B** is disposed between them in the drawings, the present disclosure is not limited thereto. For example, the first electrode **210** and the second electrode **220** may not necessarily extend in one direction and may be disposed in (or to have) various structures. For example, the first electrode **210** and the second electrode **220** may be partially curved or bent, or any one of the first electrode **210** and the second electrode **220** may surround (e.g., may extend around a periphery of) the other electrode. The structure or shape of the first electrode **210** and the second electrode **220** is not particularly limited as long as the first electrode **210** and the second electrode **220** are at least partially spaced apart to face each other so that a space in which the light-emitting elements **300** are to be disposed can be formed between the first electrode **210** and the second electrode **220**.

(46) In some embodiments, the electrode stem parts **210S** and **220S** may be omitted from the first electrode **210** and the second electrode **220**, respectively. The first electrode **210** and the second electrode **220** may only extend in one direction and may be spaced apart from each other in each subpixel PXn. This will be described later with reference to another embodiment.

(47) The banks **410**, **420**, and **430** may include the outer bank **430** disposed at a boundary between the subpixels PXn and the inner banks **410** and **420** disposed adjacent to the center of each subpixel PXn and under the electrodes **210** and **220**, respectively. A first inner bank **410** and a second inner bank **420** may be disposed under the first electrode branch part **2106** and the second electrode branch part **220B**, respectively. These will be described in more detail later with reference to other drawings.

(48) The outer bank **430** may be disposed at the boundary between the subpixels PXn. Respective ends of a plurality of first electrode stem parts **210S** may be spaced apart from each other by the outer bank **430**. The outer bank **430** may extend in the second direction DR2 to lie at the boundary between the subpixels PXn arranged in (e.g., adjacent to each other in) the first direction DR1. However, the present disclosure is not limited thereto, and the outer bank **430** may also extend in the first direction DR1 to lie at the boundary between the subpixels PXn arranged in the second direction DR2. The outer bank **430** may include the same material as the inner banks **410** and **420** and may be formed at the same time as the inner banks **410** and **420** in one process.

(49) The light-emitting elements **300** may be disposed between the first electrode **210** and the second electrode **220**. An end of each of the light-emitting elements **300** may be electrically connected to the first electrode **210**, and the other end may be electrically connected to the second electrode **220**. The light-emitting elements **300** may be electrically connected to the first electrode **210** and the second electrode **220** respectively through the contact electrodes **260**, to be described in more detail later.

(50) The light-emitting elements **300** may be spaced apart from each other and may be aligned substantially parallel to each other. A gap between the light-emitting elements **300** is not particularly limited. In some embodiments, a plurality of light-emitting elements **300** may be disposed adjacent to each other to form a group, and a plurality of other light-emitting elements **300** may form a group at a certain distance from the above group or may be oriented and aligned in a direction with non-uniform density. In an embodiment, the light-emitting elements **300** may extend in a direction, and the direction in which each electrode, for example, the first electrode branch part **210B** and the second electrode branch part **220B**, extend and the direction in which the light-emitting elements **300** extend may be substantially perpendicular to each other. However, the present disclosure is not limited thereto, and the light-emitting elements **300** may extend in a direction not perpendicular to but oblique to the direction in which the first electrode branch part **210B** and the second electrode branch part **220B** extend.

(51) The light-emitting elements **300** according to an embodiment may include the active layers **330** including different materials to emit light of different wavelength bands to the outside. The display device **10** according to an embodiment may include the light-emitting elements **300** emitting light of different wavelength bands. In the display device **10**, the light-emitting elements **300** disposed in the subpixels PXn may include the active layers **330** having the same structure but emitting (e.g., but configured to emit) different light (e.g., different color light). The light emitting elements **300** of the first subpixel PX1 may include the active layers **330** emitting first light L1 whose central wavelength band is a first wavelength. The light-emitting elements **300** of the second subpixel PX2 may include the active layers **330** emitting second light L2 whose central wavelength band is a second wavelength. The light emitting elements **300** of the third subpixel PXn may include the active layers **330** emitting third light L3 whose central wavelength band is a third wavelength.

(52) Accordingly, the first light L1 may be emitted from the first subpixel PX1, the second light L2 may be emitted from the second subpixel PX2, and the third light L3 may be emitted from the third subpixel PX3. In some embodiments, the first light L1 may be blue light having a central wavelength band in a range of about 450 to about 495 nm, the second light L2 may be green light have a central wavelength band in a range of about 495 to about 570 nm, and the third light L3 may be red light having a central wavelength band in the range of about 620 to about 752 nm.

(53) However, the present disclosure is not limited thereto. The first light L1, the second light L2, and the third light L3 may be light of different colors or light of the same color, and their central wavelength bands may also be different from the above ranges. In addition, in some embodiments, the first subpixel PX1, the second subpixel PX2, and the third subpixel PX3 may include the light-emitting elements **300** of the same type (e.g., the same light-emitting elements **300**) to emit light of substantially the same color.

(54) As illustrated in the drawings, the light-emitting elements **300** disposed in each subpixel PXn may include first light-emitting elements **301**, second light-emitting elements **302**, and third light-emitting elements **303** disposed between the first electrode branch part **210B** and the second electrode branch part **220B**. A process of manufacturing the light-emitting elements **300** may include etching a semiconductor structure **3000** (see, e.g., FIG. 11) formed on a lower substrate **1000** (see, e.g., FIG. 10) in a direction. The semiconductor structure **3000** may include an active layer **3300** having a different composition according to position, and there may be a difference in the wavelength band of light emitted from the active layer **3300**. To reduce or minimize a difference between light emitted from the light-emitting elements **300**, a composition ratio of the active layer **3300** may be controlled according to position in the semiconductor structure **3000**.

(55) A method for manufacturing the light-emitting elements **300** according to an embodiment may include forming a different pattern according to position in the active layer **3300**. Accordingly, the manufactured light-emitting elements **300** may include different patterns on their end surfaces, and positions at where the light-emitting elements **300** are formed in the semiconductor structure **3000**

may be identified by the patterns. To adjust emission characteristics of light emitted from some of the light-emitting elements **300**, composition ratios at specific positions in the semiconductor structure **3000** may be adjusted through the patterns of the light-emitting elements **300**. The display device **10** according to an embodiment may include the light-emitting elements **300** having different patterns, for example, the first light-emitting elements **301**, the second light-emitting elements **302**, and the third light-emitting elements **303** may have different patterns. The patterns of the light-emitting elements **300** will be described in more detail later.

(56) The display device **10** may include a second insulating layer **510** (see, e.g., FIG. 4), which at least partially covers the first electrode **210** and the second electrode **220**.

(57) The second insulating layer **510** may be disposed in each subpixel PXn of the display device **10**. The second insulating layer **510** may substantially cover (e.g., substantially entirely cover or entirely cover) each subpixel PXn and may also extend to other neighboring subpixels PXn. The second insulating layer **510** may at least partially cover the first electrode **210** and the second electrode **220**. The second insulating layer **510** may partially expose (e.g., may expose a portion of) the first electrode **210** and the second electrode **220**. In some embodiments, the second insulating layer **510** may partially expose the first electrode branch part **210B** and the second electrode branch part **220B**.

(58) At least a part (or portion) of each of the contact electrodes **260** may extend in a direction. The contact electrodes **260** may contact the light-emitting elements **300** and the electrodes **210** and **220**, and the light-emitting elements **300** may receive an electrical signal from the first electrode **210** and the second electrode **220** through the contact electrodes **260**.

(59) The contact electrodes **260** may include a first contact electrode **261** and a second contact electrode **262**. The first contact electrode **261** and the second contact electrode **262** may be disposed on the first electrode branch part **210B** and the second electrode branch part **220B**, respectively.

(60) The first contact electrode **261** may be disposed on the first electrode **210** or the first electrode branch part **210B** and may extend in the second direction DR2, and the first contact electrode **261** may contact an end of each of the light-emitting elements **300**. For example, the first contact electrode **261** may contact the first electrode **210** exposed without the second insulating layer **510** disposed thereon. Accordingly, the light-emitting elements **300** may be electrically connected to the first electrode **210** through the first contact electrode **261**.

(61) The second contact electrode **262** may be disposed on the second electrode **220** or the second electrode branch part **220B** and may extend in the second direction DR2. The second contact electrode **262** may be spaced apart from the first contact electrode **261** in the first direction DR1. The second contact electrode **262** may contact the other end of each of the light-emitting elements **300**. For example, the second contact electrode **262** may contact the second electrode **220** exposed without the second insulating layer **510** disposed thereon. Accordingly, the light-emitting elements **300** may be electrically connected to the second electrode **220** through the second contact electrode **262**. Although an embodiment in which two first contact electrodes **261** and one second contact electrode **262** are disposed in one subpixel PXn is illustrated in the drawings, the present disclosure is not limited thereto. The number of first contact electrodes **261** and second contact electrodes **262** may vary according to the number of first electrodes **210** and second electrodes **220** or the number of first electrode branch parts **210B** and second electrode branch parts **220B** disposed in each subpixel PXn.

(62) In some embodiments, widths of the first contact electrode **261** and the second contact electrode **262** measured in a direction may respectively be greater than widths of the first electrode **210** and the second electrode **220** or widths of the first electrode branch part **210B** and the second electrode branch part **220B** measured in the direction. Each of the first contact electrode **261** and the second contact electrode **262** may cover sides of the first electrode **210** or the second electrode **220** or sides of the first electrode branch part **210B** or the second electrode branch part **220B**.

However, the present disclosure is not limited thereto. In some embodiments, each of the first contact electrode **261** and the second contact electrode **262** may cover only one side of the first electrode branch part **2106** or the second electrode branch part **220B**.

(63) The display device **10** may include the circuit element layer PAL, which is located under each of the electrodes **210** and **220** and a third insulating layer **520** (see, e.g., FIG. 4), and a passivation layer **550** (see, e.g., FIG. 4), which at least partially cover each electrode **210** or **220** and the light-emitting elements **300**. The structure of the display device **10** according to an embodiment will now be described in more detail with reference to FIG. 4.

(64) FIG. 4 is a cross-sectional view taken along the lines Xa-Xa', Xb-Xb', and Xc-Xc' in FIG. 3.

(65) FIG. 4 illustrates a cross section of only the first subpixel PX1, but the same illustration may be applied to other pixels PX or subpixels PXn (e.g., other pixels PX or subpixel PXn may have the same or substantially similar structure as the first subpixel PX1). FIG. 4 illustrates a cross section across an end and the other end of a light-emitting element **300** disposed in the first subpixel PX1.

(66) Referring to FIG. 4, in conjunction with FIGS. 2 and 3, the display device **10** may include the circuit element layer PAL and a light emitting layer EML. The circuit element layer PAL may include a substrate **110**, a buffer layer **115**, light blocking layers BML, first and second transistors **120** and **140**, etc. The light emitting layer EML may include the plurality of electrodes **210** and **220**, the light-emitting elements **300**, a plurality of insulating layers **510**, **520** and **550**, etc., disposed on the first and second transistors **120** and **140**.

(67) The substrate **110** may be an insulating substrate. The substrate **110** may include (or may be made of) an insulating material, such as glass, quartz, or polymer resin. In addition, the substrate **110** may be a rigid substrate or may be a flexible substrate that can be bent, folded, rolled, etc.

(68) The light blocking layers BML may be disposed on the substrate **110**. The light blocking layers BML may include a first light blocking layer BML1 and a second light blocking layer BML2. The first light blocking layer BML1 may be electrically connected to a first drain electrode **123** of the first transistor **120**, which will be described in more detail later. The second light blocking layer BML2 may be electrically connected to a second drain electrode **143** of the second transistor **140**.

(69) The first light blocking layer BML1 and the second light blocking layer BML2 are overlapped by a first active material layer **126** of the first transistor **120** and a second active material layer **146** of the second transistor **140**, respectively. The first and second light blocking layers BML1 and BML2 may include a light blocking material to prevent or substantially prevent light from entering the first and second active material layers **126** and **146**. For example, the first and second light blocking layers BML1 and BML2 may include (or may be made of) an opaque metal material that blocks or substantially blocks the transmission of light. However, the present disclosure is not limited thereto. In some embodiments, the light blocking layers BML may be omitted.

(70) The buffer layer **115** is disposed on the light blocking layers BML and the substrate **110**. The buffer layer **115** may entirely cover the substrate **110** as well as the light blocking layers BML. The buffer layer **115** may prevent or substantially prevent diffusion of impurity ions, prevent or substantially prevent penetration of moisture or outside air, and perform a surface planarization function. In addition, the buffer layer **115** may insulate the light blocking layers BML and the first and second active material layers **126** and **146** from each other.

(71) A semiconductor layer is disposed on the buffer layer **115**. The semiconductor layer may include the first active material layer **126** of the first transistor **120**, the second active material layer **146** of the second transistor **140**, and an auxiliary layer **163**. The semiconductor layer may include polycrystalline silicon, monocrystalline silicon, an oxide semiconductor, or the like.

(72) The first active material layer **126** may include a first doped region **126a**, a second doped region **126b**, and a first channel region **126c**. The first channel region **126c** may be disposed between the first doped region **126a** and the second doped region **126b**. The second active material layer **146** may include a third doped region **146a**, a fourth doped region **146b**, and a second channel

region **146c**. The second channel region **146c** may be disposed between the third doped region **146a** and the fourth doped region **146b**. The first active material layer **126** and the second active material layer **146** may include polycrystalline silicon. The polycrystalline silicon may be formed by crystalizing amorphous silicon. Examples of the crystallization method include, but are not limited to, a rapid thermal annealing (RTA) method, a solid phase crystallization (SPC) method, an excimer laser annealing (ELA) method, a metal induced crystallization (MILC) method, and a sequential lateral solidification (SLS) method. In other embodiments, the first active material layer **126** and the second active material layer **146** may include monocrystalline silicon, low-temperature polycrystalline silicon, amorphous silicon, or the like. The first doped region **126a**, the second doped region **126b**, the third doped region **146a**, and the fourth doped region **146b** may be, but are not limited to, regions of the first active material layer **126** and the second active material layer **146** that are doped with impurities.

(73) The first active material layer **126** and the second active material layer **146** are not necessarily limited to the above description. In an embodiment, the first active material layer **126** and the second active material layer **146** may include an oxide semiconductor. In such an embodiment, the first doped region **126a** and the third doped region **146a** may be first conducting regions, and the second doped region **126b** and the fourth doped region **146b** may be second conducting regions. When the first active material layer **126** and the second active material layer **146** include an oxide semiconductor, the oxide semiconductor may be an oxide semiconductor including (or containing) indium (In). In some embodiments, the oxide semiconductor may be, but is not limited to, indium tin oxide (ITO), indium zinc oxide (IZO), indium gallium oxide (IGO), indium zinc tin oxide (IZTO), indium gallium tin oxide (IGTO), or indium gallium zinc tin oxide (IGZTO).

(74) A first gate insulating layer **150** is disposed on the semiconductor layer. The first gate insulating layer **150** may entirely cover the buffer layer **115** as well as the semiconductor layer. The first gate insulating layer **150** may act as a gate insulating layer for each of the first and second transistors **120** and **140**.

(75) A first conductive layer is disposed on the first gate insulating layer **150**. The first conductive layer disposed on the first gate insulating layer **150** may include a first gate electrode **121** disposed on the first active material layer **126** of the first transistor **120**, a second gate electrode **141** disposed on the second active material layer **146** of the second transistor **140**, and a power wiring **161** disposed on the auxiliary layer **163**. The first gate electrode **121** may overlap the first channel region **126c** of the first active material layer **126**, and the second gate electrode **141** may overlap the second channel region **146c** of the second active material layer **146**.

(76) An interlayer insulating film **170** is disposed on the first conductive layer. The interlayer insulating film **170** may act as an insulating film between the first conductive layer and other layers disposed on the first conductive layer. The interlayer insulating film **170** may include an organic insulating material and may perform a surface planarization function.

(77) A second conductive layer is disposed on the interlayer insulating film **170**. The second conductive layer includes the first drain electrode **123** and a first source electrode **124** of the first transistor **120**, the second drain electrode **143** and a second source electrode **144** of the second transistor **140**, and a power electrode **162** disposed on the power wiring **161**.

(78) The first drain electrode **123** and the first source electrode **124** may respectively contact the first doped region **126a** and the second doped region **126b** of the first active material layer **126** through contact openings (e.g., contact holes) penetrating (or extending through) the interlayer insulating film **170** and the first gate insulating layer **150**. The second drain electrode **143** and the second source electrode **144** may respectively contact the third doped region **146a** and the fourth doped region **146b** of the second active material layer **146** through contact openings (e.g., contact holes) penetrating the interlayer insulating film **170** and the first gate insulating layer **150**. In addition, the first drain electrode **123** and the second drain electrode **143** may be electrically connected to the first light blocking layer BML1 and the second light blocking layer BML2

through other contact openings (e.g., contact holes), respectively.

(79) A first insulating layer **200** is disposed on the second conductive layer. The first insulating layer **200** may include an organic insulating material and may perform a surface planarization function.

(80) The inner banks **410** and **420**, the outer bank **430**, the electrodes **210** and **220**, and the light-emitting elements **300** may be disposed on the first insulating layer **200**.

(81) The banks **410**, **420**, and **430** may include the inner banks **410** and **420** spaced apart from each other in each subpixel PXn and the outer bank **430** disposed at the boundary between neighboring (or adjacent) subpixels PXn.

(82) The outer bank **430** may extend in the second direction DR2 and be at (e.g., to lie at) the boundary between the subpixels PXn arranged in (e.g., adjacent to each other in) the first direction DR1. However, the present disclosure is not limited thereto, and the outer bank **430** may also extend in the first direction DR1 to be at the boundary between the subpixels PXn arranged in the second direction DR2. For example, the outer bank **430** may define the boundary of each subpixel PXn.

(83) When the display device **10** is manufacturing by a method in which ink with the light-emitting elements **300** dispersed (or suspended) therein is sprayed using an inkjet printing device, the outer bank **430** may prevent or substantially prevent the ink from flowing over (or out of) the boundary of each subpixel PXn. The outer bank **430** may separate inks in which different light-emitting elements **300** are dispersed for different subpixels PXn so that the inks do not mix with each other. However, the present disclosure is not limited thereto.

(84) The inner banks **410** and **420** may include the first inner bank **410** and the second inner bank **420** disposed adjacent to the center of each subpixel PXn.

(85) The first inner bank **410** and the second inner bank **420** are spaced apart to face each other. The first electrode **210** may be disposed on the first inner bank **410**, and the second electrode **220** may be disposed on the second inner bank **420**. As shown in FIGS. 3 and 4, the first electrode branch part **210B** is disposed on the first inner bank **410** and the second electrode branch part **220B** is disposed on the second inner bank **420**.

(86) The first inner bank **410** and the second inner bank **420** may extend in the second direction DR2 in each subpixel PXn. The first inner bank **410** and the second inner bank **420** may extend in the second direction DR2 toward a neighboring subpixel PXn in the second direction DR2. However, the present disclosure is not limited thereto, and the first inner bank **410** and the second inner bank **420** may be disposed in each subpixel PXn to form patterns in the entire display device **10**. The banks **410**, **420**, and **430** may include, but are not limited to, polyimide (PI).

(87) At least a part of each of the first inner bank **410** and the second inner bank **420** may protrude from the first insulating layer **200**. Each of the first inner bank **410** and the second inner bank **420** may protrude upwardly from a plane in which the light-emitting elements **300** are disposed, and the protruding part (or portion) may be at least partially inclined (e.g., may have one or more inclined surfaces). The protruding shape of each of the first inner bank **410** and the second inner bank **420** is not limited.

(88) When the inner banks **410** and **420** protruding from the first insulating layer **200** have inclined side surfaces, light emitted from the light-emitting elements **300** may be reflected by the inclined side surfaces of the inner banks **410** and **420**. As will be described in more detail later, when the electrodes **210** and **220** disposed on the inner banks **410** and **420** include a material having high reflectivity, light emitted from the light-emitting elements **300** may be reflected by the electrodes **210** and **220** located on the inclined side surfaces of the inner banks **410** and **420** to travel upwardly above (e.g., away from) the first insulating layer **200**.

(89) As described above, the banks **410**, **420**, and **430** may include the same material and may be formed in the same process. However, when the outer bank **430** is disposed at the boundary of each subpixel PXn to form a grid pattern, the inner banks **410** and **420** are disposed in each subpixel

PXn to extend in a direction. In addition, when the outer bank **430** separates neighboring subpixels PXn and prevents ink from overflowing to adjacent subpixels PXn in an inkjet process, the inner banks **410** and **420** have a protruding structure in each subpixel PXn to act as reflective barriers that reflect light emitted from the light-emitting elements **300** upwardly above the first insulating layer **200**. However, the present disclosure is not limited thereto.

(90) The electrodes **210** and **220** may be disposed on the first insulating layer **200** and the inner banks **410** and **420**. As described above, each of the electrodes **210** and **220** includes the electrode stem part **210S** or **220S** and the electrode branch part **210B** or **220B**. The line Xa-Xa' in FIG. **3** is a line crossing the first electrode stem part **210S**, the line Xb-Xb' in FIG. **3** is a line crossing the first electrode branch part **210B** and the second electrode branch part **220B**, and the line Xc-Xc' in FIG. **3** is a line crossing the second electrode stem part **220S**. For example, it may be understood that the first electrode **210** disposed along the area Xa-Xa' in FIG. **4** is the first electrode stem part **210S**, the first electrode **210** and the second electrode **220** disposed along the area Xb-Xb' in FIG. **4** are the first electrode branch part **210B** and the second electrode branch part **220B**, respectively, and the second electrode **220** disposed along the area Xc-Xc' in FIG. **4** is the second electrode stem part **220S**. The electrode stem parts **210S** and **220S** and the electrode branch parts **210B** and **220B** may form the first electrode **210** and the second electrode **220**, respectively.

(91) Each of the first electrode **210** and the second electrode **220** may have a part (or portion) disposed on the first insulating layer **200** and a part (or portion) disposed on the first inner bank **410** or the second inner bank **420**. As described above, the first electrode stem part **210S** of the first electrode **210** and the second electrode stem part **220S** of the second electrode **220** may extend in the first direction DR1, and the first inner bank **410** and the second inner bank **420** may extend in the second direction DR2 to also lie in a neighboring subpixel PXn in the second direction DR2. The first electrode stem part **210S** and the second electrode stem part **220S** of the first electrode **210** and the second electrode **220**, which extend in the first direction DR1, may partially overlap the first inner bank **410** and the second inner bank **420**. However, the present disclosure is not limited thereto, and the first electrode stem part **210S** and the second electrode stem part **220S** may not overlap the first inner bank **410** and the second inner bank **420**.

(92) The first electrode contact opening CNTD penetrating the first insulating layer **200** to expose a part of the first drain electrode **123** of the first transistor **120** may be formed in (or under) the first electrode stem part **210S** of the first electrode **210**. The first electrode **210** may contact the first drain electrode **123** through the first electrode contact opening CNTD. The first electrode **210** may be electrically connected to the first drain electrode **123** of the first transistor **120** to receive an electrical signal (e.g., a predetermined electrical signal).

(93) The second electrode stem part **220S** of the second electrode **220** may extend in a direction to also lie in the non-emission area in which the light-emitting elements **300** are not disposed. The second electrode contact opening CNTS penetrating the first insulating layer **200** to expose a part of the power electrode **162** may be formed in the second electrode stem part **220S**. The second electrode **220** may contact the power electrode **162** through the second electrode contact opening CNTS. The second electrode **220** may be electrically connected to the power electrode **162** to receive an electrical signal (e.g., a predetermined electrical signal) from the power electrode **162**.

(94) Parts of the first electrode **210** and the second electrode **220**, for example, the first electrode branch part **210B** and the second electrode branch part **220B**, may be disposed on the first inner bank **410** and the second inner bank **420**, respectively. The first electrode branch part **210B** of the first electrode **210** may cover the first inner bank **410**, and the second electrode branch part **220B** of the second electrode **220** may cover the second inner bank **420**. Because the first inner bank **410** and the second inner bank **420** are spaced apart from each other in the center of each subpixel PXn, the first electrode branch part **210B** and the second electrode branch part **220B** may also be spaced apart from each other. A plurality of light-emitting elements **300** may be disposed in an area between the first electrode **210** and the second electrode **220**, for example, in a space in which the

first electrode branch part **210B** and the second electrode branch part **220B** are spaced apart to face each other.

(95) Each of the electrodes **210** and **220** may include a transparent conductive material. For example, each of the electrodes **210** and **220** may include a material, such as indium tin oxide (ITO), indium zinc oxide (IZO), or indium tin zinc oxide (ITZO). However, the present disclosure is not limited thereto. In some embodiments, each of the electrodes **210** and **220** may include a conductive material having high reflectivity. For example, each of the electrodes **210** and **220** may include a metal, such as silver (Ag), copper (Cu), or aluminum (Al) as a material having high reflectivity. In such an embodiment, each of the electrodes **210** and **220** may reflect incident light in an upward direction of each subpixel PXn.

(96) In addition, each of the electrodes **210** and **220** may have a structure in which a transparent conductive material and a metal layer having high reflectivity are each stacked in one or more layers or may be formed as a single layer including them. In an embodiment, each of the electrodes **210** and **220** may have a stacked structure of ITO/Ag/ITO/IZO or may be an alloy including (or containing) aluminum (Al), nickel (Ni), lanthanum (La), etc. However, the present disclosure is not limited thereto.

(97) The second insulating layer **510** is disposed on the first insulating layer **200**, the first electrode **210**, and the second electrode **220**. The second insulating layer **510** partially covers the first electrode **210** and the second electrode **220**. The second insulating layer **510** may cover most of upper surfaces of the first electrode **210** and the second electrode **220** but may partially expose the first electrode **210** and the second electrode **220**. The second insulating layer **510** may partially expose the upper surfaces of the first electrode **210** and the second electrode **220**, for example, the second insulating layer **510** may partially expose an upper surface of the first electrode branch part **210B** disposed on the first inner bank **410** and an upper surface of the second electrode branch part **220B** disposed on the second inner bank **420**. For example, the second insulating layer **510** may be substantially entirely formed on the first insulating layer **200** but may include openings partially exposing the first electrode **210** and the second electrode **220**. The openings of the second insulating layer **510** may be positioned to expose the relatively flat upper surfaces of the first electrode **210** and the second electrode **220**.

(98) In an embodiment, the second insulating layer **510** may be stepped such that a part of an upper surface of the second insulating layer **510** is depressed between the first electrode **210** and the second electrode **220**. In some embodiments, the second insulating layer **510** may include an inorganic insulating material, and a part of the upper surface of the second insulating layer **510** disposed to cover the first electrode **210** and the second electrode **220** may be depressed due to a step formed by the members disposed under the second insulating layer **510**. The light-emitting elements **300** disposed on the second insulating layer **510** between the first electrode **210** and the second electrode **220** may form an empty space with the depressed upper surface of the second insulating layer **510**. The light-emitting elements **300** may be partially spaced apart from the upper surface of the second insulating layer **510**, and the empty space may be filled with a material that forms the third insulating layer **520** to be described later.

(99) However, the present disclosure is not limited thereto. The second insulating layer **510** may also form a flat upper surface so that the light-emitting elements **300** can be disposed on the flat upper surface. The upper surface may extend in a direction toward the first electrode **210** and the second electrode **220** and may end on inclined side surfaces of the first electrode **210** and the second electrode **220**. For example, the second insulating layer **510** may be disposed in areas where the electrodes **210** and **220** overlap the inclined side surfaces of the first inner bank **410** and the second inner bank **420**, respectively. The contact electrodes **260**, to be described in more detail later, may contact the exposed areas of the first electrode **210** and the second electrode **220** and may smoothly contact ends of the light-emitting elements **300** on the flat upper surface of the second insulating layer **510**.

(100) The second insulating layer **510** may protect the first electrode **210** and the second electrode **220** while insulating them from each other. In addition, the second insulating layer **510** may prevent the light-emitting elements **300** disposed on the second insulating layer **510** from directly contacting other members and, thus, from being damaged. However, the shape and structure of the second insulating layer **510** are not limited thereto.

(101) The light-emitting elements **300** may be disposed on the second insulating layer **510** between the electrodes **210** and **220**. For example, at least one light-emitting element **300** may be disposed on the second insulating layer **510** disposed between the electrode branch parts **210B** and **220B**. However, the present disclosure is not limited thereto. At least some of the light-emitting elements **300** disposed in each subpixel PXn may also be disposed in an area other than the area between the electrode branch parts **210B** and **220B**. The light-emitting elements **300** may be disposed at a position where a part of each light-emitting element **300** overlaps the electrodes **210** and **220**. The light-emitting elements **300** may be disposed on respective ends of the first electrode branch part **210B** and the second electrode branch part **220B** which face each other and may be electrically connected to the electrodes **210** and **220** through the contact electrodes **260**, respectively.

(102) Each of the light-emitting elements **300** may include a plurality of layers disposed in a direction horizontal to the first insulating layer **200**. Each of the light-emitting elements **300** of the display device **10** according to an embodiment may extend in one direction and may have a structure in which a plurality of semiconductor layers are sequentially disposed in the one direction. As will be described in more detail later, each of the light-emitting elements **300** may include a first semiconductor layer **310**, the active layer **330**, a second semiconductor layer **320**, and an electrode layer **370** sequentially disposed along the one direction, and an insulating film **380** may surround (e.g., may extend around) outer surfaces of the above layers. The direction in which the light-emitting elements **300** disposed in the display device **10** extend may be parallel to the first insulating layer **200**, and the semiconductor layers included in each of the light-emitting elements **300** may be sequentially disposed along a direction parallel to an upper surface of the first insulating layer **200**. However, the present disclosure is not limited thereto. In some embodiments, when the light-emitting elements **300** have a different structure, the layers therein may be disposed in a direction perpendicular to the first insulating layer **200**.

(103) In addition, an end of each of the light-emitting elements **300** may contact the first contact electrode **261**, and the other end may contact the second contact electrode **262**. According to an embodiment, the insulating film **380** may not be formed on end surfaces of each light-emitting element **300** in the extending direction of the light-emitting element **300**, thereby exposing the end surfaces. Thus, the exposed areas of each of the light-emitting elements **300** may contact the first contact electrode **261** and the second contact electrode **262**, to be described in more detail later.

(104) However, the present disclosure is not limited thereto. In some embodiments, at least a part of the insulating film **380** of each light-emitting element **300** may be removed, and the insulating film **380** may be removed (e.g., partially or selectively removed) to partially expose side surfaces of both ends of the light-emitting element **300**. In an operation of forming the third insulating layer **520** covering an outer surface of each light-emitting element **300** during a process of manufacturing the display device **10**, the insulating film **380** may be partially removed. The exposed side surfaces of each light-emitting element **300** may contact the first contact electrode **261** and the second contact electrode **262**. However, the present disclosure is not limited thereto.

(105) The third insulating layer **520** may be disposed on a part of each light-emitting element **300** disposed between the first electrode **210** and the second electrode **220**. The third insulating layer **520** may partially cover the outer surface of each light-emitting element **300**. The third insulating layer **520** may protect the light-emitting elements **300** while fixing (e.g., fixing the location of) the light-emitting elements **300** in the process of manufacturing the display device **10**. In addition, in an embodiment, some of the third insulating layer **520** may be disposed between lower surfaces of the light-emitting elements **300** and the second insulating layer **510**. As described above, the third

insulating layer **520** may be formed to fill the space between the second insulating layer **510** and each light-emitting element **300** formed during the process of manufacturing the display device **10**. Accordingly, the third insulating layer **520** may be formed to cover the outer surface of each light-emitting element **300**. However, the present disclosure is not limited thereto.

(106) The third insulating layer **520** may extend in the second direction DR2 between the first electrode branch part **2106** and the second electrode branch part **220B** in a plan view. For example, the third insulating layer **520** disposed on the first insulating layer **200** may have an island or linear shape in a plan view. According to an embodiment, the third insulating layer **520** may be disposed on the light-emitting elements **300**.

(107) Each of the first contact electrode **261** and the second contact electrode **262** is disposed on the electrode **210** or **220** and the third insulating layer **520**. The first contact electrode **261** and the second contact electrode **262** may be spaced apart from each other on the third insulating layer **520**. The third insulating layer **520** may insulate the first contact electrode **261** and the second contact electrode **262** from each other to prevent them from directly contacting each other.

(108) As described above, the first contact electrode **261** and the second contact electrode **262** may extend in the second direction DR2 in a plan view and may be spaced apart from each other in the first direction DR1. Each of the first contact electrode **261** and the second contact electrode **262** may contact at least one end of each light-emitting element **300** and may be electrically connected to the first electrode **210** or the second electrode **220** to receive an electrical signal.

(109) The first contact electrode **261** may contact an exposed area of the first electrode **210** on the first inner bank **410**, and the second contact electrode **262** may contact an exposed area of the second electrode **220** on the second inner bank **420**.

(110) The first contact electrode **261** and the second contact electrode **262** may send an electrical signal received from each electrode **210** or **220** to the light-emitting elements **300**.

(111) The contact electrodes **260** may include a conductive material, such as ITO, IZO, ITZO, or aluminum (Al). However, the present disclosure is not limited thereto.

(112) The passivation layer **550** may be disposed on the contact electrodes **260** and the third insulating layer **520**. The passivation layer **550** may protect the members disposed on the first insulating layer **200** from the external environment.

(113) Each of the second insulating layer **510**, the third insulating layer **520**, and the passivation layer **550** described above may include an inorganic insulating material or an organic insulating material. In an embodiment, the second insulating layer **510**, the third insulating layer **520**, and the passivation layer **550** may include an inorganic insulating material, such as silicon oxide (SiO₂), silicon nitride (SiN_x), silicon oxynitride (SiO_xN_y), aluminum oxide (Al₂O₃), or aluminum nitride (AlN). In other embodiments, the second insulating layer **510**, the third insulating layer **520**, and the passivation layer **550** may include an organic insulating material, such as acrylic resin, epoxy resin, phenolic resin, polyamide resin, polyimide resin, unsaturated polyester resin, polyphenylene resin, polyphenylene sulfide resin, benzocyclobutene, cardo resin, siloxane resin, silsesquioxane resin, polymethyl methacrylate, polycarbonate, or polymethyl methacrylate-polycarbonate synthetic resin. However, the present disclosure is not limited thereto.

(114) The display device **10** may include a greater number of insulating layers. According to an embodiment, the display device **10** may further include a fourth insulating layer **530** disposed to protect the first contact electrode **261**.

(115) FIG. 5 is a cross-sectional view of a part of a display device according to an embodiment.

(116) Referring to FIG. 5, a display device **10** according to an embodiment may further include a fourth insulating layer **530** disposed on a first contact electrode **261**. The display device **10** illustrated in FIG. 5 is different from the display device **10** illustrated in FIG. 4 in that it further includes the fourth insulating layer **530**, and thus, at least a part of a second contact electrode **262** is disposed on the fourth insulating layer **530**. Therefore, any redundant description therebetween

may be omitted and differences will be mainly described below.

(117) The display device **10** shown in FIG. **5** may include the fourth insulating layer **530**, which is disposed on the first contact electrode **261** and electrically insulates the first contact electrode **261** and the second contact electrode **262** from each other. The fourth insulating layer **530** may cover the first contact electrode **261** but may not overlap a part (or portion) of each light-emitting element **300** so that the light-emitting element **300** can be connected to the second contact electrode **262**. The fourth insulating layer **530** may be disposed on an upper surface of a third insulating layer **520** to partially contact the first contact electrode **261** and the third insulating layer **520**. The fourth insulating layer **530** may cover an end of the first contact electrode **261** on the third insulating layer **520**. Accordingly, the fourth insulating layer **530** may protect the first contact electrode **261** while electrically insulating the first contact electrode **261** from the second contact electrode **262**.

(118) A side surface of the fourth insulating layer **530** in a direction in which the second contact electrode **262** is disposed may be aligned with a side surface of the third insulating layer **520**. However, the present disclosure is not limited thereto. In some embodiments, the fourth insulating layer **530** may include an inorganic insulating material, similar to or the same as the second insulating layer **510**.

(119) The first contact electrode **261** may be disposed between a first electrode **210** and the fourth insulating layer **530**, and the second contact electrode **262** may be disposed on the fourth insulating layer **530**. The second contact electrode **262** may partially contact the second insulating layer **510**, the third insulating layer **520**, the fourth insulating layer **530**, a second electrode **220**, and the light-emitting elements **300**. An end of the second contact electrode **262** in a direction in which the first electrode **210** is disposed may be disposed on the fourth insulating layer **530**.

(120) A passivation layer **550** may be disposed on the fourth insulating layer **530** and the second contact electrode **262** to protect them. A redundant description thereof may be omitted.

(121) As described above, the display device **10** according to an embodiment may include the light-emitting elements **300**, each having a pattern formed on an end surface thereof. The display device **10** may include a plurality of light-emitting elements **300** having different patterns, for example, the first light-emitting elements **301**, the second light-emitting elements **302**, and the third light-emitting elements **303**. Each of the first light-emitting elements **301**, the second light-emitting elements **302**, and the third light-emitting elements **303** includes a pattern part **370P** (see, e.g., FIG. **7**) formed on an end surface thereof. At which position in the semiconductor structure **3000** each light-emitting element **300** is manufactured can be identified by the pattern part **370P** formed on the light-emitting element **300**. A light-emitting element **300** according to an embodiment will now be described in more detail with reference to other drawings.

(122) FIG. **6** is a schematic view of a light-emitting element according to an embodiment, and FIG. **7** is a cross-sectional view taken along the line VI-VI' in FIG. **6**.

(123) A light-emitting element **300** may be a light emitting diode. For example, the light-emitting element **300** may be an inorganic light emitting diode having a size of micrometers or nanometers and including (or made of) an inorganic material. When an electric field is formed in a specific direction between two electrodes facing each other, the inorganic light emitting diode may be aligned between the two electrodes in which polarities are formed. The light-emitting element **300** may be aligned between two electrodes by an electric field formed between the two electrodes.

(124) The light-emitting element **300** according to an embodiment may extend in one direction. The light-emitting element **300** may have a rod shape, a wire shape, a tube shape, or the like. In an embodiment, the light-emitting element **300** may have a cylindrical or a rod shape. However, the shape of the light-emitting element **300** is not limited thereto, and the light-emitting element **300** may have various suitable shapes including polygonal prisms, such as a cube, a rectangular parallelepiped, and a hexagonal prism, and a shape extending in a direction and having a partially inclined outer surface. A plurality of semiconductors included in the light-emitting element **300**, which will be described in more detail later, may be sequentially disposed or stacked along the one

direction.

(125) The light-emitting element **300** may include a semiconductor layer doped with impurities of any conductivity type (e.g., a p-type or an n-type). The semiconductor layer may receive an electrical signal from an external power source and emit light in a specific wavelength band.

(126) The light-emitting element **300** according to an embodiment may emit light in a specific wavelength band. In an embodiment, the active layer **330** may emit blue light having a central wavelength band is in a range of about 450 to about 495 nm. However, the central wavelength band of the blue light is not limited to the above range, and it should be understood to include all wavelength ranges that can be recognized as blue in the art to which the present disclosure pertains. In addition, light emitted from the active layer **330** of the light-emitting element **300** is not limited to the blue light and may also be green light having a central wavelength band in a range of about 495 to about 570 nm or red light having a central wavelength band in a range of about 620 to about 750 nm. The light-emitting element **300** emitting the blue light will be described in more detail below as an example.

(127) Referring to FIGS. **6** and **7**, the light-emitting element **300** may include a first semiconductor layer **310**, a second semiconductor layer **320**, the active layer **330**, an electrode layer **370**, and the insulating film **380**.

(128) The first semiconductor layer **310** may be an n-type semiconductor. In an example, when the light-emitting element **300** emits light in a blue wavelength band, the first semiconductor layer **310** may include a semiconductor material having a chemical formula of $\text{Al}_{0.5x}\text{Ga}_{0.5y}\text{In}_{0.5(1-x-y)}\text{N}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$), for example, any one or more of n-type doped AlGaInN, GaN, AlGaIn, InGaIn, AlN, and InN. The first semiconductor layer **310** may be doped with an n-type dopant, and the n-type dopant may be, for example, Si, Ge, or Sn. In an embodiment, the first semiconductor layer **310** may be n-GaN doped with n-type Si. A length of the first semiconductor layer **310** may be in a range of, but is not limited to, about 1.5 to about 5 μm .

(129) The second semiconductor layer **320** is disposed on the active layer **330** to be described in more detail later. The second semiconductor layer **320** may be a p-type semiconductor. In an example, when the light-emitting element **300** emits light in a blue or green wavelength band, the second semiconductor layer **320** may include a semiconductor material having a chemical formula of $\text{Al}_{0.5x}\text{Ga}_{0.5y}\text{In}_{0.5(1-x-y)}\text{N}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$), for example, any one or more of p-type doped AlGaInN, GaN, AlGaIn, InGaIn, AlN, and InN. The second semiconductor layer **320** may be doped with a p-type dopant, and the p-type dopant may be, for example, Mg, Zn, Ca, Se, or Ba. In an embodiment, the second semiconductor layer **320** may be p-GaN doped with p-type Mg. A length of the second semiconductor layer **320** may be in a range of, but is not limited to, about 0.05 to about 0.10 μm .

(130) Although each of the first semiconductor layer **310** and the second semiconductor layer **320** is illustrated as being composed of one layer in the drawings, the present disclosure is not limited thereto. According to some embodiments, each of the first semiconductor layer **310** and the second semiconductor layer **320** may include more layers, for example, a clad layer or a tensile strain barrier reducing layer depending on the material of the active layer **330**. This will be described in more detail later with reference to other drawings.

(131) The active layer **330** is disposed between the first semiconductor layer **310** and the second semiconductor layer **320**. The active layer **330** may include a material having a single or multiple quantum well structure. When the active layer **330** includes a material having a multiple quantum well structure, it may have a structure in which a plurality of quantum layers and a plurality of well layers are alternately stacked. The active layer **330** may emit light through combination of electron-hole pairs according to an electrical signal received through the first semiconductor layer **310** and the second semiconductor layer **320**. For example, when the active layer **330** emits light in the blue wavelength band, it may include a material such as AlGaIn or AlGaInN. When the active layer **330** has a multiple quantum well structure in which a quantum layer and a well layer are alternately

stacked, the quantum layer may include a material such as AlGaInN or AlGaInN, and the well layer may include a material such as GaN or AlInN. In an embodiment, the active layer **330** may include AlGaInN as a quantum layer and AlInN as a well layer to emit blue light having a central wavelength band in a range of about 450 to about 495 nm as described above.

(132) However, the present disclosure is not limited thereto, and the active layer **330** may also have a structure in which a semiconductor material having a large band gap energy and a semiconductor material having a small band gap energy are alternately stacked or may include different group 3 to 5 semiconductor materials depending on the wavelength band of light that it emits. Light emitted from the active layer **330** is not limited to light in the blue wavelength band. In some embodiments, the active layer **330** may emit light in a red or green wavelength band. A length of the active layer **330** may be in a range of, but is not limited to, about 0.05 to about 0.10 μm .

(133) Light emitted from the active layer **330** may be radiated not only to an outer surface of the light-emitting element **300** in a longitudinal direction but also to both side surfaces. The direction of light emitted from the active layer **330** is not limited to one direction.

(134) The electrode layer **370** may be an ohmic contact electrode. However, the present disclosure is not limited thereto, and the electrode layer **370** may also be a Schottky contact electrode. The light-emitting element **300** may include at least one electrode layer **370**. Although the light-emitting element **300** illustrated in FIG. 6 includes one electrode layer **370**, the present disclosure is not limited thereto. In some embodiments, the light-emitting element **300** may include more electrode layers **370** or the electrode layer **370** may be omitted. The following description of the light-emitting element **300** may apply equally even if the number of electrode layers **370** is changed or another structure is further included.

(135) When the light-emitting element **300** is electrically connected to an electrode or a contact electrode in a display device **10** according to an embodiment, the electrode layer **370** may reduce the resistance between the light-emitting element **300** and the electrode or the contact electrode. The electrode layer **370** may include a conductive metal. For example, the electrode layer **370** may include at least any one of aluminum (Al), titanium (Ti), indium (In), gold (Au), silver (Ag), indium tin oxide (ITO), indium zinc oxide (IZO), and indium tin zinc oxide (ITZO). In addition, the electrode layer **370** may include an n-type or p-type doped semiconductor material. The electrode layer **370** may include the same material or different materials, but the present disclosure is not limited thereto.

(136) The insulating film **380** surrounds (e.g., extends around a periphery of) outer surfaces of the semiconductor layers and electrode layers described above. In an embodiment, the insulating film **380** may surround an outer surface of at least the active layer **330** and extend in the direction in which the light-emitting element **300** extends. The insulating film **380** may protect the above members. For example, the insulating film **380** may surround side surfaces of the above members but may expose both ends (e.g., opposite ends) of the light-emitting element **300** in the longitudinal direction.

(137) The insulating film **380** is illustrated as extending in the longitudinal direction of the light-emitting element **300** to cover from side surfaces of the first semiconductor layer **310** to side surfaces of the electrode layer **370**. However, the present disclosure is not limited thereto, and the insulating film **380** may also cover outer surfaces of only some semiconductor layers as well as the active layer **330** or may cover only a part of an outer surface of the electrode layer **370** to partially expose the outer surface of the electrode layer **370**. In addition, an upper surface of the insulating film **380** may be rounded in a cross-sectional view in an area adjacent to at least one end of the light-emitting element **300**.

(138) A thickness of the insulating film **380** may be in a range of, but is not limited to, about 10 nm to about 1.0 μm . In one embodiment, the thickness of the insulating film **380** may be about 40 nm.

(139) The insulating film **380** may include a material having insulating properties, such as silicon oxide ($\text{SiO}_{\text{sub.x}}$), silicon nitride ($\text{SiN}_{\text{sub.x}}$), silicon oxynitride ($\text{SiO}_{\text{sub.x}}\text{N}_{\text{sub.y}}$), aluminum

nitride (AlN), or aluminum oxide (Al.sub.2O.sub.3). Therefore, the insulating film **380** may prevent or substantially prevent an electrical short circuit that may otherwise occur if the active layer **330** directly contacted an electrode through which an electrical signal is transmitted to the light-emitting element **300**. In addition, the insulating film **380** may prevent or substantially prevent a reduction in luminous efficiency by protecting the outer surface of the light-emitting element **300** including the active layer **330**.

(140) In some embodiments, an outer surface of the insulating film **380** may be treated. When the display device **10** is manufactured, the light-emitting element **300** may be sprayed on electrodes in a state in which the light-emitting element **300** is dispersed in an ink, and may then be aligned. Here, the surface of the insulating film **380** may be hydrophobic or hydrophilic-treated so that the light-emitting element **300** remains separate from other adjacent light-emitting elements **300** in the ink and does not agglomerate with the other light-emitting elements **300**.

(141) A length *h* of the light-emitting element **300** may be in a range of about 1 to about 10 μm or about 2 to about 6 μm and, in some embodiments, may be in a range of about 3 to about 5 μm . In addition, a diameter of the light-emitting element **300** may be in a range of about 300 to about 700 nm, and an aspect ratio of the light-emitting element **300** may be in a range of about 1.2 to about 100. However, the present disclosure is not limited thereto, and a plurality of light-emitting elements **300** included in the display device **10** may have different diameters according to a difference in composition of the active layer **330**. The diameter of the light-emitting element **300** may be about 500 nm.

(142) The light-emitting element **300** according to an embodiment may include a protrusion pattern **370P** formed on a surface of the electrode layer **370**. The protrusion pattern **370P** may be formed by partially etching an upper surface of the electrode layer **370** during a process of manufacturing the light-emitting element **300**. Although the protrusion pattern **370P** is shown in FIGS. **6** and **7** as being formed on the exposed upper surface of the electrode layer **370**, the present disclosure is not limited thereto. In some embodiments, the protrusion pattern **370P** may be formed by partially etching an upper surface of the second semiconductor layer **320**, and the exposed upper surface of the electrode layer **370** may form a flat surface. An embodiment where the protrusion pattern **370P** is formed on the upper surface of the electrode layer **370** will be described below as an example.

(143) The protrusion pattern **370P** may include a plurality of protrusions protruding from a part (or portion) of the upper surface of the electrode layer **370** and a depression other than (e.g., between) the protrusions. The depression is an area between the protrusions and may be an area depressed from the protrusions.

(144) According to an embodiment, the protrusion pattern **370P** may include an first protrusion, a second protrusion spaced apart from the first protrusion, and a depression formed between them. The protrusion pattern **370P** may include one or more protrusions, and the protrusions may be spaced apart from each other. As illustrated in the drawings, the protrusion pattern **370P** may include any one protrusion and other protrusions surrounding (e.g., extending around a periphery of) the protrusion, and a depression may be formed between them. However, the present disclosure is not limited thereto, and the protrusions of the protrusion pattern **370P** may also be formed as a plurality of patterns spaced apart from each other. For example, any one protrusion may not surround another protrusion but may be spaced apart from the protrusion to face the protrusion.

(145) In some embodiments, the protrusions of the protrusion pattern **370P** may have curved outer surfaces and may have a protruding shape. As illustrated in FIG. **6**, the protrusion pattern **370P** may include a circular first protrusion which is located in the center of the upper surface of the electrode layer **370** and circular second and third protrusions which are spaced apart from the first protrusion to surround (e.g., to be concentric with) the first protrusion. However, the present disclosure is not limited thereto, and the protrusions of the protrusion pattern **370P** may also extend in a direction or may have a mesh structure. In addition, the number of protrusions included in the protrusion pattern **370P** of the light-emitting element **300** may vary.

(146) According to an embodiment, the protrusions may have the same width and may be spaced apart at regular intervals. A width of the first protrusion may be equal to widths of the second protrusion and the third protrusion, and a distance between the first protrusion and the second protrusion and a distance between the second protrusion and the third protrusion may be equal to each other. The widths of the protrusions of the protrusion pattern **370P** and the distance between the protrusions may vary according to the shape of a second pattern layer **1800** (see, e.g., FIG. **21**) formed to etch the upper surface of the electrode layer **370** during the processing of the light-emitting element **300**, as will be described in more detail later. For example, the number, width, and spacing of the protrusions included in the protrusion pattern **370P** may vary according to the shape of the second pattern layer **1800**. However, the protrusions do not necessarily have the same width and spacing. Only some of the protrusions may have the same width as another protrusions or may be spaced apart from another protrusion by the same distance between other protrusion.

(147) In addition, according to an embodiment, the protrusion pattern **370P** of the light-emitting element **300** may be formed on the upper surface of the electrode layer **370**, and both end surfaces of the light-emitting element **300** may have different roughness. As will be described in more detail later, the protrusion pattern **370P** may be formed by partially etching a surface of the semiconductor structure **3000**. Accordingly, in the semiconductor structure **3000**, a surface on which the protrusion pattern **370P** is formed may form an uneven surface, but the other surface may have a flat surface. The protrusion pattern **370P** may be formed on an end surface of the light-emitting element **300** formed by etching the semiconductor structure **3000**, and the other end surface of the light-emitting element **300** may have a flat surface. In other words, both end surfaces of the light-emitting element **300** may have different roughness. In the specification, “roughness” indicates that the flatness of one surface is relatively different from that of the other surface. For example, the light-emitting element **300** according to an embodiment may include an end surface on which the protrusion pattern **370P** is formed and the other end surface on which the protrusion pattern **370P** is not formed. Because the flatness of the end surface is lower than that of the other end surface, the end surfaces may be understood to have different roughness. However, the present disclosure is not limited thereto, and it may be understood the same way in an embodiment different from the above as long as it is within a range that can be interpreted as having meaning of the term “roughness.”

(148) The light-emitting element **300** may include both end surfaces having different roughness and may be electrically connected to different electrodes **210** and **220**, respectively. As illustrated in FIG. **4**, according to an embodiment, an end surface of the light-emitting element **300** on which the protrusion pattern **370P** is formed may be electrically connected to the first electrode **210**, and the other end surface may be electrically connected to the second electrode **220**. The first contact electrode **261** may contact the first electrode **210** and the protrusion pattern **370P** of the light-emitting element **300**, and the second contact electrode **262** may contact the second electrode **220** and the first semiconductor layer **310** of the light-emitting element **300**. However, the present disclosure is not limited thereto. This described above, and thus, a repeated description thereof will be omitted.

(149) The protrusion pattern **370P** of the light-emitting element **300** may be formed by an etching process performed during the process of manufacturing the light-emitting element **300**. The process of manufacturing the light-emitting element **300** includes a process of etching the semiconductor structure **3000** (see, e.g., FIG. **11**) in a direction, and light-emitting characteristics of the manufactured light-emitting element **300** may vary according to position in the semiconductor structure **3000**. The semiconductor structure **3000** manufactured by growing a semiconductor material may have a different composition according to position. Accordingly, light emitted from light-emitting elements **300** manufactured may have different emission values (or amounts) or different central wavelength bands. To manufacture light-emitting elements **300** having uniform light-emitting characteristics, a mark (e.g., a predetermined mark or indicator) may be used to

identify the position at which each light-emitting element **300** having certain light-emitting characteristics is manufactured in the semiconductor structure **3000**.

(150) Because the light-emitting element **300** according to an embodiment includes the protrusion pattern **370P** formed on a surface of the electrode layer **370**, the position at which the light-emitting element **300** having certain light-emitting characteristics is formed in the semiconductor structure **3000** can be identified during the process of manufacturing the light-emitting element **300**.

Accordingly, the composition ratio of the semiconductor material can be controlled according to position in the semiconductor structure **3000**, and light-emitting elements **300** having uniform light-emitting characteristics can be manufactured. The display device **10** according to an embodiment may include a plurality of light-emitting elements **300** having uniform light-emitting characteristics but different protrusion patterns **370P**.

(151) FIG. **8** is a cross-sectional view taken along the line VII-VII' in FIG. **3**.

(152) Referring to FIG. **8**, the display device **10** according to an embodiment may include a first light-emitting element **301**, a second light-emitting element **302**, and a third light-emitting element **303** having different protrusion patterns **370P**. The first light-emitting element **301**, the second light-emitting element **302**, and the third light-emitting element **303** may each have a protrusion pattern **370P** including protrusions and depressions and may each include a different number of protrusions. The first light-emitting element **301** may include a first protrusion pattern **371P** having three protrusions and two depressions formed between them. The second light-emitting element **302** may include a second protrusion pattern **372P** having two protrusions and one depression. The third light-emitting element **303** may include a third protrusion pattern **373P** having one protrusion and one depression. In FIG. **8**, each of the first light-emitting element **301**, the second light-emitting element **302**, and the third light-emitting element **303** includes circular protrusions, and any one protrusion surrounds another protrusion. However, the present disclosure is not limited thereto, and the protrusions of the protrusion patterns **370P** may have various shapes as will be described in more detail later.

(153) The light-emitting elements **300** having different protrusion patterns **370P** may be formed at different positions in the semiconductor structure **3000** during a manufacturing process. The position at which each light-emitting element **300** is formed in the semiconductor structure **3000** may be identified according to the shape of the protrusion pattern **370P** formed in the light-emitting element **300**, the number of protrusions, and the like. For example, based on light-emitting characteristics of the first light-emitting element **301** including the first protrusion pattern **371P**, the second light-emitting element **302** and the third light-emitting element **303** may have different light-emitting characteristics from the first light-emitting element **301**. To control their light-emitting characteristics, semiconductor compositions at positions where the second protrusion pattern **372P** and the third protrusion pattern **373P** of the second light-emitting element **302** and the third light-emitting element **303** are formed in the semiconductor structure **300** may be adjusted. By adjusting composition ratios in areas in which the second protrusion pattern **372P** and the third protrusion pattern **373P** are located based on an area in which the first protrusion pattern **371P** of the semiconductor structure **3000** is located, the first light-emitting element **301**, the second light-emitting element **302**, and the third light-emitting element **303** may be made to have uniform or substantially uniform light-emitting characteristics. Accordingly, the display device **10** according to an embodiment may include the light-emitting elements **300** including different protrusion patterns **370P** but having uniform or substantially uniform light-emitting characteristics, and each subpixel PXn may have uniform or substantially uniform light-emitting characteristics by including various light-emitting elements **300**.

(154) A process of manufacturing light-emitting elements **300** according to an embodiment will now be described.

(155) FIG. **9** is a flowchart illustrating a method for manufacturing light-emitting elements according to an embodiment.

(156) Referring to FIG. 9, the method for manufacturing the light-emitting elements **300** according to an embodiment may include providing a semiconductor structure **3000** formed on a substrate (operation **S100**) and forming a plurality of element rods **ROD**, each including a protrusion pattern **370P** protruding from at least a part of an upper surface, by etching the semiconductor structure **3000**, with different protrusion patterns **370P** being formed in different areas **WA** defined in the semiconductor structure **3000** (operation **S200**).

(157) As described above, the light-emitting elements **300** may be manufactured through a process of etching the semiconductor structure **3000**. The method for manufacturing the light-emitting elements **300** according to an embodiment may include a process of forming the protrusion patterns **370P** to indicate positions at which the light-emitting elements **300** are formed in the semiconductor structure **3000**. The manufactured light-emitting elements **300** may include the protrusion patterns **370P** described above, and positions having different light-emitting characteristics in the semiconductor structure **3000** can be identified by using the protrusions **370P**. The method for manufacturing the light-emitting elements **300** will now be described in more detail with further reference to other drawings.

(158) FIGS. **10** through **25** are schematic views illustrating a process of manufacturing light-emitting elements according to an embodiment.

(159) First, referring to FIG. **10**, a lower substrate **1000** which includes a base substrate **1100** and a buffer material layer **1200** formed on the base substrate **1100** is prepared. The base substrate **1100** may be (or may include) a sapphire substrate (Al.sub.2O.sub.3) or a transparent substrate, such as glass. However, the present disclosure is not limited thereto, and the base substrate **1100** may also be a conductive substrate, such as GaN, SiC, ZnO, Si, GaP, or GaAs. An embodiment in which the base substrate **1100** is a sapphire substrate (Al.sub.2O.sub.3) will be described below as an example. A thickness of the base substrate **1100** is not particularly limited but may be, for example, in a range of about 400 to about 1500 μm .

(160) A plurality of semiconductor layers are formed on the base substrate **1100**. The semiconductor layers grown by an epitaxial method may be formed by growing seed crystals. The method for forming the semiconductor layers may be electron beam deposition, physical vapor deposition (PVD), chemical vapor deposition (CVD), plasma laser deposition (PLD), dual-type thermal evaporation, sputtering, or metal organic chemical vapor deposition (MOCVD). In one embodiment, the method for forming the semiconductor layers may be MOCVD, but the present disclosure is not limited thereto.

(161) A precursor material for forming the semiconductor layers is not particularly limited within a range of suitable materials that can be generally selected to form a target material. For example, the precursor material may include a metal precursor including an alkyl group, such as a methyl group or an ethyl group. For example, the metal precursor may be, but is not limited to, a compound such as trimethyl gallium (Ga(CH.sub.3).sub.3), trimethyl aluminum (Al(CH.sub.3).sub.3), or triethyl phosphate (C.sub.2H.sub.5).sub.3PO.sub.4. The method or process conditions for forming the semiconductor layers will not be described below, but the order of the method for manufacturing the light-emitting elements **300** and the stacked structure of each light-emitting element **300** will be described in detail.

(162) The buffer material layer **1200** is formed on the base substrate **1100**. Although the buffer material layer **1200** is illustrated as a single layer in the drawings, the present disclosure is not limited thereto, and a plurality of layers may also be formed. The buffer material layer **1200** may be disposed to reduce a difference in lattice constant between a first semiconductor **3100** and the base substrate **1100**.

(163) For example, the buffer material layer **1200** may include an undoped semiconductor and may include substantially the same material as the first semiconductor **3100** but may be a material not doped with an n type or a p type. In an embodiment, the buffer material layer **1200** may be, but is not limited to, at least any one of undoped InAlGaN, GaN, AlGaN, InGaN, AlN, and InN. In

addition, the buffer material layer **1200** may be omitted depending on the base substrate **1100**. An embodiment in which the buffer material layer **1200** including an undoped semiconductor is formed on the base substrate **1100** will be described as an example.

(164) Next, referring to FIG. **11**, a semiconductor structure **3000** is formed on the lower substrate **1000**. The semiconductor structure **3000** may include the first semiconductor **3100**, an active layer **3300**, a second semiconductor **3200**, and an electrode material layer **3700**. A plurality of material layers included in the semiconductor structure **3000** may be formed by performing a conventional process as described above. A plurality of layers included in the semiconductor structure **3000** may correspond to a plurality of layers of each light-emitting element **300** according to an embodiment, respectively. For example, the layers may include the same materials as a first semiconductor layer **310**, an active layer **330**, a second semiconductor layer **320**, and an electrode layer **370** of each light-emitting element **300**, respectively.

(165) As illustrated in FIG. **11**, the active layer **3300** of the semiconductor structure **3000** may include areas WA having different light-emitting characteristics at different positions. The areas WA may be defined at different positions in the semiconductor structure **3000**.

(166) Depending on the performance or quality of a manufacturing device, the semiconductor structure **3000** formed on the lower substrate **1000** may have an area having a partially non-uniform composition according to a spatial position. For example, when the semiconductor structure **3000** is formed through MOCVD, a precursor material provided on the lower substrate **1000** may be in a gas phase. The precursor material in the gas phase may be provided on the lower substrate **1000** in a non-uniform distribution. Accordingly, the semiconductor structure **3000** formed by depositing precursor materials may have a non-uniform composition according to a spatial position.

(167) Here, when the active layer **3300** of the semiconductor structure **3000** has a different composition according to a spatial position, the respective active layers **330** of a plurality of light-emitting elements **300** manufactured may have different compositions. The light-emitting elements **300** manufactured at different positions in the semiconductor structure **3000** may have different light-emitting characteristics, for example, different light emission amounts and central wavelength bands of emitted light. The semiconductor structure **3000** may include a first area WA1, a second area WA2, and a third area WA3. In some embodiments, the first area WA1, the second area WA2, and the third area WA3 may be distinguished according to light-emitting characteristics of the active layer **3300** of the semiconductor structure **3000**. The second area WA2 and the third WA3 may have light emission values (or amounts) or central wavelength bands of light different from those of the first area WA1. However, the present disclosure is not limited thereto, and the first area WA1, the second area WA2, and the third area WA3 of the semiconductor structure **3000** may also be arbitrarily defined areas.

(168) The method for manufacturing the light-emitting elements **300** according to an embodiment may include forming a different protrusion pattern **370P** on an upper surface of each area WA. Therefore, the position at which each light-emitting element **300** is formed in the semiconductor structure **3000** can be easily identified. When light-emitting characteristics of a light-emitting element **300** having a specific protrusion pattern **370P** are different from those of other light-emitting elements **300**, the composition of the active layer **3300** in an area WA in which the protrusion pattern **370P** is formed in the semiconductor structure **3000** may be adjusted.

(169) The method for manufacturing the light-emitting elements **300** according to an embodiment may include forming a first pattern layer **1700** for forming element rods ROD and forming a second pattern layer **1800** for forming the protrusion patterns **370P**. The semiconductor structure **3000** etched along mask patterns of the first pattern layer **1700** may form semiconductor crystals **3000'**, each including the first semiconductor layer **310**, the active layer **330**, the second semiconductor layer **320**, etc. The semiconductor structure **3000** etched along mask patterns of the second pattern layer **1800** may form the protrusion patterns **370P**.

(170) A method for forming the semiconductor crystals **3000'** includes forming an etch mask layer **1600** and the first pattern layer **1700** on the semiconductor structure **3000** and forming the semiconductor crystals **3000'** by etching the semiconductor structure **3000** in a direction perpendicular to the lower substrate **1000**.

(171) First, as illustrated in FIGS. **12** through **15**, the etch mask layer **1600** and the first pattern layer **1700** are formed on the semiconductor structure **3000**. The etch mask layer **1600** is formed on the electrode material layer **3700**, and the first pattern layer **1700** is formed on the etch mask layer **1600**. The etch mask layer **1600** may act as a mask for successively etching the layers of the semiconductor structure **3000**. The etch mask layer **1600** may include a first etch mask layer **1610** including an insulating material and a second etch mask layer **1620** including a metal.

(172) The first etch mask layer **1610** may include oxide or nitride as an insulating material. The insulating material may be, for example, silicon oxide (SiO.sub.x), silicon nitride (SiN.sub.x), or silicon oxynitride (SiO.sub.xN.sub.y). A thickness of the first etch mask layer **1610** may be in a range of, but is not limited to, about 0.5 to about 1.5 μm .

(173) The second etch mask layer **1620** is disposed on the first etch mask layer **1620**. For example, the second etch mask layer **1620** may be a hard mask layer. The second etch mask layer **1620** may include a material that acts as a mask for successive etching of the semiconductor structure **3000** and may include, for example, a metal, such as chromium (Cr). A thickness of the second etch mask layer **1620** may be in a range of, but is not limited to, about 30 to about 150 nm.

(174) The first pattern layer **1700** may be disposed on the etch mask layer **1600**. The first pattern layer **1700** may include first through third mask patterns **1710**, **1720**, and **1730** spaced apart from each other to act as a mask for successive etching of the semiconductor structure **3000**. The first pattern layer **1700** may include a polymer, polystyrene spheres, silica spheres, or the like but is not limited to a particular material as long as it is a suitable material that can form patterns.

(175) For example, when the first pattern layer **1700** includes a polymer, a conventional method for forming patterns using the polymer may be used. For example, the first pattern layer **1700** including the polymer may be formed by using a method such as photolithography, e-beam lithography, or nano-imprint lithography.

(176) In an embodiment, the first pattern layer **1700** may be formed by nano-imprint lithography. As illustrated in the drawings, the forming of the first pattern layer **1700** may include spraying (e.g., depositing) a first resin ink Ink1 on the etch mask layer **1600** by using a first head Head1 and forming the first through third mask patterns **1710**, **1720**, and **1730** by curing a first resin **1700'**. The mask patterns of the first pattern layer **1700** may include a nano-imprint resin. The resin may include, but is not limited to, a fluorinated monomer, an acrylate monomer, dipentaerythritol hexaacrylate, dipropylene glycol diacrylate, poly(ethylene glycol) phenyl ether acrylate, butylated hydroxy toluene (BHT), or 1-hydroxy-cyclohexylphenylketone (also known as Irgacure 184).

(177) The first pattern layer **1700** may include the first mask patterns **1710** formed in the first area WA1, the second mask patterns **1720** formed in the second area WA2, and the third mask pattern **1730** formed in the third area WA3. The first mask patterns **1710**, the second mask patterns **1720**, and the third mask pattern **1730** may be spaced apart from each other.

(178) FIG. **15** is a plan view of the semiconductor structure **3000** on which the first pattern layer **1700** is formed viewed from above. As illustrated in FIG. **15**, the semiconductor structure **3000** may include the first area WA1, the second area WA2, and the third area WA3 in which the active layer **3300** having different light-emitting characteristics is located. In the drawing, the first area WA1 surrounds (e.g., extends around a periphery of) the second area WA2, and the second area WA2 surrounds (e.g., extends around a periphery of) the third area WA3. However, the present disclosure is not limited thereto.

(179) The first mask patterns **1710**, the second mask patterns **1720**, and the third mask pattern **1730** of the first pattern layer **1700** may be spaced apart from each other and may be formed in the first area WA1, the second area WA2, and the third area WA3, respectively. A gap between them and

their shapes may vary according to the shapes of the light-emitting elements **300** to be manufactured. In some embodiments, the first mask patterns **1710**, the second mask patterns **1720**, and the third mask pattern **1730** may have the same diameter or width. For example, a first width **m1** of the first mask patterns **1710** may be equal to a second width **m2** of the second mask patterns **1720** and a third width **m3** of the third mask pattern **1730**. In a subsequent process, the semiconductor structure **3000** is etched along the gaps between the first through third mask patterns **1710**, **1720**, and **1730** of the first pattern layer **1700**. Therefore, the shapes of the first through third mask patterns **1710**, **1720**, and **1730** may be substantially similar to the cross-sectional shapes of the light-emitting elements **300**. Because the first through third mask patterns **1710**, **1720**, and **1730** have the same diameter or width, the light-emitting elements **300** may have the same width.

(180) Next, referring to FIGS. **16** and **17**, the semiconductor structure **3000** is etched along the first pattern layer **1700** to form the semiconductor crystals **3000'**. The forming of the semiconductor crystals **3000'** may include a first etching operation for forming first etch holes (e.g., first etch grooves) Hole1 by etching the etch mask layer **1600** and the electrode material layer **3700** in the direction perpendicular to the lower substrate **1000** in areas between the first through third mask patterns **1710**, **1720**, and **1730** of the first pattern layer **1700**, a second etching operation for forming second etch holes (e.g., second etch grooves) Hole2 by etching from the second semiconductor **3200** to the first semiconductor **3100** along the first etch holes Hole1, and an operation of removing the first pattern layer **1700** and the etch mask layer **1600**.

(181) The process of etching the semiconductor structure **3000** may be performed by using a conventional method. For example, the etching process may be dry etching, wet etching, reactive ion etching (RIE), inductively coupled plasma reactive ion etching (ICP-RIE), or the like. Dry etching may be suitable for vertical etching because anisotropic etching is possible. When the above etching methods are used, an etchant may be, but is not limited to, Cl.sub.2 or O.sub.2.

(182) In some embodiments, the semiconductor structure **3000** may be etched by using a combination of dry etching and wet etching. For example, etching in a depth direction may be performed by using dry etching. Then, wet etching, which is isotropic etching, may be performed so that etched sidewalls lie in a plane perpendicular to the surface.

(183) The first etching operation for forming the first etch holes Hole1 by etching the etch mask layer **1600** and the electrode material layer **3700** along the areas between the first through third mask patterns **1710**, **1720**, and **1730** of the first pattern layer **1700** is performed. The first etch holes Hole1 may be formed by removing the first pattern layer **1700**, the etch mask layer **1600**, and the electrode material layer **3700** and may partially expose the second semiconductor **3200**. Next, the second etching operation for forming the second etch holes Hole2 by etching from the second semiconductor **3200** to the first semiconductor **3100** along (e.g., within) the first etch holes Hole1 is performed. Although the first etching operation and the second etching operation are illustrated in the drawings as being performed in separate processes, the present disclosure is not limited thereto. The first etching operation and the second etching operation may also be performed successively in one process. Each of the semiconductor crystals **3000'** formed by the etching of the semiconductor structure **3000** may include the first semiconductor layer **310**, the active layer **330**, the second semiconductor layer **320**, and the electrode layer **370**, such as in each light-emitting element **300** according to an embodiment described above.

(184) Next, the semiconductor crystals **3000'** are formed by removing the etch mask layer **1600** and the first pattern layer **1700**. The removing of the etch mask layer **1600** or the first pattern layer **1700** may be performed by a conventional process. For example, the process may be, but is not limited to, RIE, or ICP-RIE.

(185) Next, an insulating film **380** is formed to partially surround an outer surface of each of the semiconductor crystals **3000'**.

(186) Referring to FIGS. **18** and **19**, the insulating films **380** may be formed by forming an insulating film **3800** surrounding the outer surfaces of the semiconductor crystals **3000'** and then

partially removing the insulating film **3800** to expose upper surfaces of the electrode layers **370**. Because the insulating film **3800** is formed on the outer surfaces of the semiconductor crystals **3000'** including the electrode layers **370**, the insulating films **380** of the light-emitting elements **300** may also be formed on outer surfaces of the electrode layers **370**.

(187) The insulating film **3800** is an insulating material formed on the outer surfaces of the semiconductor crystals **3000'** and may be formed by coating or immersing an insulating material on the outer surfaces of the vertically etched semiconductor crystals **3000'**. However, the present disclosure is not limited thereto. For example, the insulating film **3800** may be formed by atomic layer deposition (ALD).

(188) The insulating film **3800** may be formed not only on side surfaces and upper surfaces of the semiconductor crystals **3000'** but also on the lower substrate **1000** exposed between the semiconductor crystals **3000'**, which are spaced apart from each other. The insulating film **3800** may be partially removed to expose the upper surfaces of the electrode layers **370**, thereby forming the insulating films **380**. The process of partially removing the insulating film **3800** may be achieved by a process such as dry etching, which is anisotropic etching, or etch-back. In the drawings, an upper surface of the insulating film **3800** is removed to expose the electrode layers **370**, and upper surfaces of the insulating films **380** are flat. However, the present disclosure is not limited thereto. In some embodiments, an outer surface of each insulating film **380** may be partially curved in an area around the electrode layer **370**. Because not only the upper surface but also side surfaces of the insulating film **3800** are partially removed in the process of partially removing the insulating film **3800**, each insulating film **380** surrounding a plurality of layers may be formed such that an end surface thereof is partially etched. For example, when the upper surface of the insulating film **3800** is removed, the outer surface of the insulating film **380** adjacent to the electrode layer **370** in each light-emitting element **300** may be partially removed.

(189) Next, the second pattern layer **1800** is formed on the semiconductor crystals **3000'** on which the insulating films **380** are formed, and the element rods ROD, each including the protrusion pattern **370P** protruding from at least a part of an upper surface thereof, are formed. According to an embodiment, the forming of the element rods ROD may include forming the second pattern layer **1800** including fourth through sixth mask patterns **1810**, **1820**, and **1830** spaced apart from each other and forming the protrusion patterns **370P** by partially etching the upper surfaces of the semiconductor crystals **3000'** along an area in which the fourth through sixth mask patterns **1810**, **1820**, and **1830** are spaced apart from each other.

(190) As illustrated in FIGS. **20** and **21**, the second pattern layer **1800** is formed on the semiconductor crystals **3000'** on which the insulating films **380** are formed. The second pattern layer **1800** may include the fourth through sixth mask patterns **1810**, **1820**, and **1830** spaced apart from each other, that is, the fourth mask patterns **1810**, the fifth mask patterns **1820**, and the sixth mask pattern **1830**. The fourth through sixth mask patterns **1810**, **1820**, and **1830** of the second pattern layer **1800** may be formed of substantially the same material as the first through third mask patterns **1710**, **1720**, and **1730** of the first pattern layer **1700** through the same or substantially similar process. As illustrated in the drawings, the forming of the second pattern layer **1800** may include spraying (e.g., depositing) a second resin ink Ink2 on the semiconductor crystals **3000'** by using a second head Head2 and forming the fourth through sixth mask patterns **1810**, **1820**, and **1830** by curing a second resin **1800'**. This is the same as described above, and thus, a repeated description thereof will be omitted.

(191) The second pattern layer **1800** may include the fourth mask patterns **1810** formed on semiconductor crystals **3000'** at a position overlapping the first area WA1, the fifth mask patterns **1820** formed on semiconductor crystals **3000'** at a position overlapping the second area WA2, and the sixth mask pattern **1830** formed on a semiconductor crystal **3000'** at a position overlapping the third area WA3. The fourth mask patterns **1810**, the fifth mask patterns **1820**, and the sixth mask pattern **1830** may be spaced apart from each other.

(192) FIG. 22 is a plan view of the semiconductor crystals **3000'** on which the second pattern layer **1800** is formed viewed from above. Referring to FIGS. 21 and 22, the second pattern layer **1800** according to an embodiment may include the fourth through sixth mask patterns **1810**, **1820**, and **1830**, which are different from each other. The fourth mask patterns **1810**, the fifth mask patterns **1820**, and the sixth mask pattern **1830** may each include a plurality of pattern parts and may each include a different number of pattern parts. The pattern parts may be spaced apart from each other to form each of the fourth through sixth mask patterns **1810**, **1820**, and **1830**. In the drawing, the pattern parts of each of the fourth through sixth mask patterns **1810**, **1820**, and **1830** are formed such that any one pattern part surrounds (e.g., extends around) another pattern part. However, the present disclosure is not limited thereto. The shapes of the fourth through sixth mask patterns **1810**, **1820**, and **1830** may vary according to the shapes of the protrusion patterns **370P** to be formed on the light-emitting elements **300**.

(193) Each of the fourth mask patterns **1810** formed on semiconductor crystals **3000'** at a position overlapping the first area **WA1** may include three pattern parts. Each of the fifth mask patterns **1820** formed on semiconductor crystals **3000'** at a position overlapping the second area **WA2** may include two pattern parts, and the sixth mask pattern **1830** formed on a semiconductor crystal **3000'** at a position overlapping the third area **WA3** may include one pattern part.

(194) Referring to FIGS. 23 and 24, the upper surfaces of the semiconductor crystals **3000'** are partially etched along the fourth through sixth mask patterns **1810**, **1820**, and **1830** of the second pattern layer **1800** to form the element rods **ROD** having the protrusion patterns **370P**.

(195) As described above, the method for manufacturing the light-emitting elements **300** according to an embodiment includes forming the protrusion patterns **370P** on the semiconductor crystals **3000'**. In the forming of the protrusion patterns **370P**, the fourth through sixth mask patterns **1810**, **1820**, and **1830** of the second pattern layer **1800** may have different shapes, and the manufactured light-emitting elements **300** may include different protrusion patterns **370P** in different areas **WA** of the semiconductor structure **3000**.

(196) For example, because the fourth mask patterns **1810** are formed in the first area **WA1**, light-emitting elements **300** formed in the first area **WA1** may be first light-emitting elements **301** having the first protrusion pattern **371P** shown in FIG. 8. Because the fifth mask patterns **1820** are formed in the second area **WA2**, light-emitting elements **300** formed in the second area **WA2** may be second light-emitting elements **302** having the second protrusion pattern **372P** shown in FIG. 8. Because the sixth mask pattern **1830** is formed in the third area **WA3**, a light-emitting element **300** formed in the third area **WA3** may be a third light-emitting element **303** having the third protrusion pattern **373P** shown in FIG. 8.

(197) The light-emitting elements **300** manufactured through the semiconductor structure **3000** may include different protrusion patterns **370P** in different areas **WA** defined in the semiconductor structure **3000**. When there is a difference between the light-emitting characteristics of the manufactured light-emitting elements **300**, an area **WA** in which a corresponding light-emitting element **300** is formed in the semiconductor structure **3000** may be tracked (e.g., identified) by the protrusion pattern **370P** of the light-emitting element **300**, and the light-emitting characteristics of the area **WA** in the semiconductor structure **3000** may be adjusted. Accordingly, the light-emitting elements **300** manufactured on the lower substrate **1000** can have uniform light-emitting characteristics.

(198) Finally, as illustrated in FIG. 25, the element rods **ROD** are separated from the lower substrate **1000** to manufacture the light-emitting elements **300**. The manufactured light-emitting elements **300** may include different protrusion patterns **370P** in different areas **WA** of the semiconductor structure **3000**. The light-emitting elements **300** may include the first light-emitting elements **301**, the second light-emitting elements **302**, and the third light-emitting element **303** having the protrusion patterns **370P** with different shapes.

(199) The light-emitting elements **300** according to an embodiment may be manufactured through

the process described above. The method for manufacturing the light-emitting elements **300** may include defining a plurality of areas WA in the semiconductor structure **3000**, etching the semiconductor structure **3000** according to the first pattern layer **1700**, and forming the protrusion patterns **370P** according to the second pattern layer **1800**. The first through third mask patterns **1710**, **1720**, and **1730** of the first pattern layer **1700** may have the same shape even though they are disposed in different areas WA of the semiconductor structure **3000**, but the fourth through sixth mask patterns **1810**, **1820**, and **1830** of the second pattern layer **1800** may have different shapes in different areas WA. Accordingly, the manufactured light-emitting elements **300** may have the same shape or diameter in different areas WA of the semiconductor structure **3000**, but the shapes of the protrusion patterns **370P** may be different. The position at which each light-emitting element **300** is formed in the semiconductor structure **3000** can be identified by the shape of the protrusion pattern **370P** of the light-emitting element **300**. Because the light-emitting elements **300** are manufactured to have different protrusion patterns **370P** in different areas defined in the semiconductor structure **3000**, in which area of the semiconductor structure **3000** a light-emitting element **300** is manufactured can be identified through the protrusion pattern **370P** of the light-emitting element **300**, and light-emitting characteristics of light emitted from the area WA of the semiconductor structure **3000** can be adjusted. Accordingly, the light-emitting elements **300** having uniform light-emitting characteristics can be manufactured by repeating the manufacturing process several times, and a display device **10** including the light-emitting elements **300** having different protrusion patterns **370P** can have uniform light-emitting characteristics in each subpixel PXn.

(200) Methods of manufacturing light-emitting elements **300** and light-emitting elements **300** according to various embodiments will now be described.

(201) FIGS. **26** through **28** are plan views illustrating shapes of a second pattern layer according to an embodiment.

(202) During a process of manufacturing light-emitting elements **300**, shapes of the fourth through sixth mask patterns **1810**, **1820**, and **1830** of a second pattern layer **1800** are not limited to those illustrated in FIG. **22**.

(203) First, referring to FIG. **26**, fourth through sixth mask patterns **1810_1**, **1820_1**, and **1830_1** of a second pattern layer **1800_1** according to an embodiment may each include a plurality of pattern parts spaced apart from each other. The embodiment illustrated in FIG. **26** is different from the embodiment illustrated in FIG. **22** in shapes of the fourth through sixth mask patterns **1810_1**, **1820_1**, and **1830_1**. Therefore, any redundant description will be omitted, and differences therebetween will be primarily described below.

(204) In the second pattern layer **1800_1** shown in FIG. **26**, pattern parts of the fourth through sixth mask patterns **1810_1**, **1820_1**, and **1830_1** may have a shape not in which any one pattern part surrounds another pattern part but in which a plurality of pattern parts are spaced apart from each other. However, the fourth through sixth mask patterns **1810_1**, **1820_1**, and **1830_1** may include different numbers of pattern parts in different areas WA in which the second pattern layer **1800_1** is formed.

(205) Each of the fourth mask patterns **1810_1** formed in a first area WA1 may include four different pattern parts spaced apart from one pattern part in vertical and horizontal directions (as shown in the drawings). Each of the fifth mask patterns **1820_1** formed in a second area WA2 may include fourth patterns without the pattern part located in the center of each fourth mask pattern **1810_1**. The sixth mask pattern **1830_1** formed in a third area WA3 may include only one pattern part located in the center of each fourth mask pattern **1810_1**.

(206) Shapes of protrusion patterns **370P** of light-emitting elements **300** manufactured may vary according to shapes of the fourth through sixth mask patterns **1810_1**, **1820_1**, and **1830_1**. First light-emitting element **301** having first protrusion patterns **371P** according to the fourth mask patterns **1810_1** may each include five protrusions, second light-emitting elements **302** having second protrusion patterns **372P** according to the fifth mask patterns **1820_1** may each include four

protrusions, and a third light-emitting element **303** having a third protrusion pattern **373P** according to the sixth mask pattern **1830_1** may include one protrusion. According to an embodiment, the fourth through sixth mask patterns **1810_1**, **1820_1**, and **1830_1** of the second pattern layer **1800_1** are not particularly limited in shape and number and can be variously modified as long as they are provided in different shapes and numbers in different areas WA defined in a semiconductor structure **3000**. The first light-emitting elements **301**, the second light-emitting elements **302**, and the third light-emitting element **303** may include different protrusions **370P** according to the shapes of the fourth through sixth mask patterns **1810_1**, **1820_1**, and **1830_1** of the second pattern layer **1800_1**. Accordingly, the areas WA in which they are formed in the semiconductor structure **3000** can be tracked (e.g., identified).

(207) Next, referring to FIG. 27, fourth through sixth mask patterns **1810_2**, **1820_2**, and **1830_2** of a second pattern layer **1800_2** according to an embodiment may include pattern parts having different shapes. The embodiment illustrated in FIG. 27 is different from the embodiment illustrated in FIG. 26 in shapes of the fourth through sixth mask patterns **1810_2**, **1820_2**, and **1830_2**. Therefore, any redundant description will be omitted, and differences therebetween will be primarily described below. In the second pattern layer **1800_2** shown in FIG. 27, any one of the pattern parts of each of the fourth through sixth mask patterns **1810_2**, **1820_2**, and **1830_2** may have a shape other than a circular shape. Although any one of the pattern parts of each of the fourth through sixth mask patterns **1810_2**, **1820_2**, and **1830_2** has a different shape in the drawing, the present disclosure is not limited thereto. The fourth through sixth mask patterns **1810_2**, **1820_2**, and **1830_2** may include pattern parts having different shapes in different areas WA in which the second pattern layer **1800_2** is formed. The second pattern layer **1800_2** according to the embodiment illustrated in FIG. 27 is different from the embodiment illustrated in FIG. 26 in that the fourth through sixth mask patterns **1810_2**, **1820_2**, and **1830_2** are distinguished not by the number of pattern parts but by the shape of the pattern parts. Other details are substantially the same, and thus, a detailed description thereof will be omitted.

(208) In the above embodiments, the areas WA defined in the semiconductor structure **3000** change from the center toward the periphery of the semiconductor structure **3000**. In such an embodiment, the fourth through sixth mask patterns **1810**, **1820**, and **1830** of the second pattern layer **1800** may include different pattern parts from the center toward the periphery of the semiconductor structure **3000**. However, the shapes of the areas WA defined in the semiconductor structure **3000** are not necessarily limited thereto.

(209) Referring to FIG. 28, areas WA defined in a semiconductor structure **3000** may change from a side toward the other side of the semiconductor structure **3000**. The embodiment illustrated in FIG. 28 is different from the embodiment illustrated in FIG. 22 in positions of fourth through sixth mask patterns **1810_3**, **1820_3**, and **1830_3** of a second pattern layer **1800_3** due to changed positions of the areas WA defined in the semiconductor structure **3000**. In FIG. 28, shapes of pattern parts of the fourth through sixth mask patterns **1810_3**, **1820_3**, and **1830_3** are not illustrated. However, it is understood that the shapes of the pattern parts such as described in connection with the embodiments shown in FIGS. 22, 26, and 27 can be used. Therefore, any redundant description will be omitted, and differences therebetween will be primarily described below.

(210) As described above, when the semiconductor structure **3000** is formed by depositing a precursor material on a lower substrate **1000**, the distribution of the precursor material provided on the lower substrate **1000** may not be uniform. The arrangement of the areas WA defined in the semiconductor structure **3000** may vary according to the uniformity of the precursor material distribution.

(211) In the embodiment illustrated in FIG. 28, a first area WA may be formed on a side of the semiconductor structure **3000**, and a second area WA2 and a third area WA3 may be sequentially formed toward the other side opposite the above side. Accordingly, the arrangement of the fourth

through sixth mask patterns **1810_3**, **1820_3**, and **1830_3** of the second pattern layer **1800_3** may also be changed.

(212) The fourth mask patterns **1810_3**, the fifth mask patterns **1820_3**, and the sixth mask patterns **1830_3** may be formed in the first area **WA1**, the second area **WA2**, and the third area **WA3**, respectively, and the shape of the second pattern layer **1800_3** may change from one side toward the other side along an axial direction crossing the center of the semiconductor structure **3000**. However, the present disclosure is not limited thereto, and the opposite may also be true. The fourth through sixth mask patterns **1810_3**, **1820_3**, and **1830_3** of the second pattern layer **1800_2** may have various shapes, arrangement structures, diameters, etc., in different areas **WA** defined in the semiconductor structure **3000**.

(213) FIGS. **29** and **30** are cross-sectional views illustrating a part of a process of manufacturing light-emitting elements according to an embodiment.

(214) According to an embodiment, a second pattern layer **1800** for forming protrusion patterns **370P** may not necessarily be formed on semiconductor crystals **3000'** on which insulating films **380** are formed. In some embodiments, the second pattern layer **1800** may be formed on a semiconductor structure **3000** or may be formed on the semiconductor crystals **3000'** before the insulating films **380** are formed.

(215) Referring to FIG. **29**, a second pattern layer **1800_4** according to an embodiment may be formed on a semiconductor structure **3000** before a first pattern layer **1700** is formed. The embodiment illustrated in FIG. **29** is different in that the order of forming the first pattern layer **1700** and the second pattern layer **1800_4** is reversed. In an embodiment, a method for manufacturing light-emitting elements **300** may include forming the second pattern layer **1800_4** on the semiconductor structure **3000**, forming protrusion patterns **370P** along the second pattern layer **1800_4**, forming the first pattern layer **1700** on the semiconductor structure **3000**, and forming element rods **ROD** by etching the semiconductor structure **3000** along the first pattern layer **1700**. For example, the second pattern layer **1800_4** may be formed before the semiconductor structure **3000** is etched in a direction perpendicular to a lower substrate **1000**, and the semiconductor structure **3000** may be etched in a state where the protrusion patterns **370P** are formed, thereby forming semiconductor crystals **3000'**. The current embodiment is the same as described above except for the order of the process of forming the second pattern layer **1800_4**, and thus, a detailed description thereof will be omitted.

(216) Next, referring to FIG. **30**, according to an embodiment, a second pattern layer **1800** may be formed before insulating films **380** surrounding outer circumferential surfaces of semiconductor crystals **3000'** are formed. The embodiment illustrated in FIG. **30** is different from the embodiment illustrated in FIG. **20** in that a second resin ink **Ink2** for forming the second pattern layer **1800** is sprayed (e.g., deposited) on the semiconductor crystals **3000'** on which the insulating films **380** are not formed. Other details are the same, and thus, a detailed description thereof will be omitted.

(217) According to an embodiment, the second pattern layer **1800** for forming protrusion patterns **370P** may be formed before an electrode material layer **3700** of a semiconductor structure **3000** is formed. For example, the protrusion patterns **370P** of light-emitting elements **300** may be formed on surfaces of electrode layers **370** which contact second semiconductor layers **320**.

(218) FIGS. **31a** and **31b** are cross-sectional views of a light-emitting element according to an embodiment.

(219) Referring to FIGS. **31a** and **31b**, a light-emitting element **300_6** according to an embodiment may include a second protrusion pattern **320P_6** formed on a contact surface between an electrode layer **370_6** and a second semiconductor layer **320_6**, and the second protrusion pattern **320P_6** may contact the second semiconductor layer **320_6**. In the light-emitting element **300_6** shown in FIG. **31a**, the second protrusion pattern **320P_6** is formed only between the electrode layer **370_6** and the second semiconductor layer **320_6**. In the light-emitting element **300_6** shown in FIG. **31b**, a first protrusion pattern **370P_6** is formed in addition to the second protrusion pattern **320P_6**. The

current embodiment is different from the embodiment illustrated in FIG. 7 in the position of the protrusion pattern **320P_6** and the shapes of the second semiconductor layer **320_6** and the electrode layer **370_6**. Therefore, any redundant description will be omitted, and differences therebetween will be primarily described below.

(220) In the light-emitting element **300_6** shown in FIGS. **31a** and **31b**, the second protrusion pattern **320P_6** may be further formed between the second semiconductor layer **320_6** and the electrode layer **370_6** by performing a process of forming a second pattern layer **1800** on a second semiconductor **3200** of a semiconductor structure **3000** during a process of manufacturing the light-emitting element **300_6**. The second protrusion pattern **320P_6** may include protrusions protruding from at least a part of an upper surface of the second semiconductor layer **320_6**, and a surface of the electrode layer **370_6** may be at least partially depressed to contact the protrusions of the second protrusion pattern **320P_6**. Accordingly, both end surfaces of the light-emitting element **300_6** may form substantially flat surfaces.

(221) As illustrated in FIG. **31a**, even if a protrusion pattern **370P** is not formed on an end surface of a light-emitting element **300**, the shape of the protrusion pattern **370P** can be identified through a cross section of the light-emitting element **300** having different light-emitting characteristics. Therefore, an area **WA** of the semiconductor structure **3000** in which the light-emitting element **300** is formed can be identified, and light-emitting characteristics of the area **WA** can be adjusted. However, the present disclosure is not limited thereto, and the light-emitting element **300_6** may also include a plurality of protrusion patterns, for example, the first protrusion pattern **370P_6** and the second protrusion pattern **320P_6**.

(222) The structure of the light-emitting element **300** is not limited to that illustrated in FIG. **6**, and the light-emitting element **300** may also have other structures.

(223) FIG. **32** is a schematic view of a light-emitting element according to an embodiment, and FIG. **33** is a cross-sectional view taken along the line VI-VI' in FIG. **32**.

(224) Referring to FIGS. **32** and **33**, a light-emitting element **300'_7** according to an embodiment may further include a third semiconductor layer **330'_7**, which is disposed between a first semiconductor layer **310'_7** and an active layer **360'_7**, and a fourth semiconductor layer **340'_7** and a fifth semiconductor layer **350'_7**, which are disposed between the active layer **360'_7** and a second semiconductor layer **320'_7**. The light-emitting element **300'_7** shown in FIGS. **32** and **33** is different from the embodiment illustrated in FIG. **6** in that a plurality of semiconductor layers **330'_7**, **340'_7**, and **350'_7** and an electrode layer **372'_7** are further disposed and that the active layer **360'_7** includes (or contains) a different element. The arrangement and structure of an insulating film **380'_7** are substantially the same as those shown in FIG. **6**. In FIG. **32**, some members are given new reference numerals for ease of description even though they are the same as those shown in FIG. **6**. Any redundant description will be omitted, and differences therebetween will be primarily described below.

(225) As described above, the light-emitting element **300** shown in FIG. **6** may emit blue or green light when the active layer **330** includes nitrogen (N). On the other hand, in the light-emitting element **300'_7** shown in FIGS. **32** and **33**, each of the active layer **360'_7** and other semiconductor layers may be a semiconductor including at least phosphorous (P). For example, the light-emitting element **300'_7** according to an embodiment may emit red light having a central wavelength band in a range of about 620 to about 750 nm. However, the central wavelength band of the red light is not limited to the above range and should be understood to include all wavelength ranges that can be recognized as red in the art to which the present disclosure pertains.

(226) In some embodiments, the first semiconductor layer **310'_7** may be an n-type semiconductor layer. When the light-emitting element **300'_7** emits red light, the first semiconductor layer **310'_7** may include a semiconductor material having a chemical formula of $\text{In}_{\text{sub.x}}\text{Al}_{\text{sub.y}}\text{Ga}_{\text{sub.1-x-y}}\text{P}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$). For example, the first semiconductor layer **310'_7** may be any one or more of n-type doped InAlGaP, GaP, AlGaP, InGaP, AlP, and InP. The first semiconductor layer

310'₇ may be doped with an n-type dopant, and the n-type dopant may be, for example, Si, Ge, or Sn. In an embodiment, the first semiconductor layer **310'**₇ may be n-AlGaInP doped with n-type Si. A length of the first semiconductor layer **310'**₇ may be in a range of, but is not limited to, about 1.5 to about 5 μm .

(227) The second semiconductor layer **320'**₇ may be a p-type semiconductor layer. When the light-emitting element **300'**₇ emits red light, the second semiconductor layer **320'**₇ may include a semiconductor material having a chemical formula of $\text{In}_{0.5x}\text{Al}_{0.5y}\text{Ga}_{1-x-y}\text{P}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$). For example, the second semiconductor layer **320'**₇ may be any one or more of p-type doped InAlGaP, GaP, AlGaNP, InGaP, AlP, and InP. The second semiconductor layer **320'**₇ may be doped with a p-type dopant, and the p-type dopant may be, for example, Mg, Zn, Ca, Se, or Ba. In an embodiment, the second semiconductor layer **320'**₇ may be p-GaP doped with p-type Mg. A length of the second semiconductor layer **320'**₇ may be in a range of, but is not limited to, about 0.08 to about 0.25 μm .

(228) The active layer **360'**₇ may be disposed between the first semiconductor layer **310'**₇ and the second semiconductor layer **320'**₇. Similar to the active layer **330** shown in FIG. 6, the active layer **360'**₇ shown in FIGS. 32 and 33 may include a material having a single or multiple quantum well structure to emit light in a specific wavelength band. For example, when the active layer **360'**₇ emits light in a red wavelength band, the active layer **360'**₇ may include a material such as AlGaP or AlInGaP. For example, when the active layer **360'**₇ has a multiple quantum well structure in which a quantum layer and a well layer are alternately stacked, the quantum layer may include a material such as AlGaP or AlInGaP, and the well layer may include a material such as GaP or AlInP. In an embodiment, the active layer **360'**₇ may include AlGaInP as a quantum layer and AlInP as a well layer to emit red light having a central wavelength band in a range of about 620 to about 750 nm.

(229) The light-emitting element **300'**₇ shown in FIGS. 32 and 33 may include a clad layer disposed adjacent to the active layer **360'**₇. As illustrated in the drawings, the third semiconductor layer **330'**₇ and the fourth semiconductor layer **340'**₇ disposed between the first semiconductor layer **310'**₇ and the second semiconductor layer **320'**₇ above and below the active layer **360'**₇ may be clad layers.

(230) The third semiconductor layer **330'**₇ may be disposed between the first semiconductor layer **310'**₇ and the active layer **360'**₇. Similar to the first semiconductor layer **310'**₇, the third semiconductor layer **330'**₇ may be an n-type semiconductor. For example, the third semiconductor layer **330'**₇ may include a semiconductor material having a chemical formula of $\text{In}_{0.5x}\text{Al}_{0.5y}\text{Ga}_{1-x-y}\text{P}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$). In an embodiment, the first semiconductor layer **310'**₇ may be n-AlGaInP, and the third semiconductor layer **330'**₇ may be n-AlInP. However, the present disclosure is not limited thereto.

(231) The fourth semiconductor layer **340'**₇ may be disposed between the active layer **360'**₇ and the second semiconductor layer **320'**₇. Similar to the second semiconductor layer **320'**₇, the fourth semiconductor layer **340'**₇ may be an n-type semiconductor. For example, the fourth semiconductor layer **340'**₇ may include a semiconductor material having a chemical formula of $\text{In}_{0.5x}\text{Al}_{0.5y}\text{Ga}_{1-x-y}\text{P}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$). In an embodiment, the second semiconductor layer **320'**₇ may be p-GaP, and the fourth semiconductor layer **340'**₇ may be p-AlInP.

(232) The fifth semiconductor layer **350'**₇ may be disposed between the fourth semiconductor layer **340'**₇ and the second semiconductor layer **320'**₇. Similar to the second semiconductor layer **320'**₇ and the fourth semiconductor layer **340'**₇, the fifth semiconductor layer **350'**₇ may be a p-type doped semiconductor. In some embodiments, the fifth semiconductor layer **350'**₇ may reduce a difference in lattice constant between the fourth semiconductor layer **340'**₇ and the second semiconductor layer **320'**₇. For example, the fifth semiconductor layer **350'**₇ may be a tensile strain barrier reducing layer. For example, the fifth semiconductor layer **350'**₇ may include, but is

not limited to, p-GaInP, p-AlInP, or p-AlGaInP.

(233) A first electrode layer **371'**₇ and a second electrode layer **372'**₇ may be disposed on the first semiconductor layer **310'**₇ and the second semiconductor layer **320'**₇, respectively, different from the light-emitting element **300** shown in FIG. **6**. The first electrode layer **371'**₇ may be disposed on a lower surface of the first semiconductor layer **310'**₇, and the second electrode layer **372'**₇ may be disposed on an upper surface of the second semiconductor layer **320'**₇. However, the present disclosure is not limited thereto, and at least any one of the first electrode layer **371'**₇ and the second electrode layer **372'**₇ may be omitted. For example, in the light-emitting element **300'**₇, the first electrode layer **371'**₇ may not be disposed on the lower surface of the first semiconductor layer **310'**₇ as in the light-emitting element **300** shown in FIG. **6**, and only one second electrode layer **372'**₇ may be disposed on the upper surface of the second semiconductor layer **320'**₇.

(234) The light-emitting element **300'**₇ shown in FIGS. **32** and **33** may also be manufactured in substantially the same manner as the light-emitting element **300** shown in FIG. **6**. For example, the light-emitting element **300'**₇ shown in FIGS. **32** and **33** may be formed by forming a semiconductor structure **3000** including a plurality of semiconductor layers and the active layer **360'**₇ and forming a protrusion pattern **370P'**₇ through a second pattern layer **1800**. However, the light-emitting element **300'**₇ according to the current embodiment is different in that the semiconductor structure **3000** includes a greater number of electrode material layers and semiconductors. This is substantially the same as described above, and thus, a detailed description thereof will be omitted.

(235) According to some embodiments, the electrode stem parts **210S** and **220S** extending in the first direction DR1 may be omitted from the first electrode **210** and the second electrode **220**.

(236) FIG. **34** is a plan view of a subpixel of a display device according to an embodiment.

(237) Referring to FIG. **34**, in a display device **10_8**, first electrodes **210_8** and second electrodes **220_8** may extend in a direction, for example, in the second direction DR2. The first electrodes **210_8** and the second electrodes **220_8** may not include electrode stem parts **210S** and **220S** extending in the first direction DR1. The display device **10_8** shown in FIG. **34** is different from the display device **10** shown in FIG. **3** in that it does not include the electrode stem parts **210S** and **220S** and includes one more second electrode **220_8**. A cross section taken along the lines Xa-Xa', Xb-Xb', and Xc-Xc' in FIG. **34** may be substantially the same as FIG. **4**. Any redundant description will be omitted, and differences therebetween will be primarily described below.

(238) As illustrated in FIG. **34**, a plurality of first electrodes **210_8** and a plurality of second electrodes **220_8** may extend in the second direction DR2 in each subpixel PXn. An outer bank **430** may also extend in the second direction DR2. The second electrodes **220_8** and the outer bank **430** may also extend to other neighboring subpixels PXn in the second direction DR2. Accordingly, the subpixels PXn neighboring in the second direction DR2 may receive the same electrical signal from the second electrodes **220_8**.

(239) Different from the display device **10** shown in FIG. **3**, in the display device **10_8** shown in FIG. **34**, a second electrode contact opening (e.g., a second electrode contact hole) CNTS may be disposed in (or under) each of the second electrodes **220_8**. Each of the second electrodes **220** may be electrically connected to a power electrode **162** of a circuit element layer PAL through the second electrode contact opening CNTS located in each subpixel PXn. Although the second electrode contact opening CNTS is formed in (or under) each of the two second electrodes **220_8** in the drawing, the present disclosure is not limited thereto.

(240) The first electrodes **210_8** may extend in the second direction DR2 but may end at the boundary of each subpixel PXn. The subpixels PXn neighboring in the second direction DR2 may each include the first electrodes **210_8** spaced apart from each other and may receive different electrical signals through first electrode contact openings (e.g., first electrode contact holes) CNTD. This shape of the first electrodes **210_8** may be formed by forming the first electrodes **210_8**

extending in the second direction DR2 and then cutting the first electrodes **210_8** at the boundary between neighboring subpixels PXn during a process of manufacturing the display device **10**. In the embodiment shown in FIG. **34**, light-emitting elements **300** between one first electrode **210_8** and one second electrode **220_8** may be connected in parallel to light-emitting elements **300** between another first electrode **210_8** and another second electrode **220_8**.

(241) In the display device **10_8** shown in FIG. **34**, some of the electrodes **210_8** and **220_8** may be floating electrodes not electrically connected to the circuit element layer PAL through the electrode contact openings CNTD and CNTS. For example, only outermost electrodes from among the electrodes **210_8** and **220_8** may receive an electrical signal through the electrode contact openings CNTD and CNTS, and electrodes **210_8** and **220_8** disposed between them may not directly receive an electrical signal. In such an embodiment, some of the second electrodes **220_8**, for example, the second electrode **220_8** disposed between the different first electrodes **210_8**, may extend in the second direction DR2 but, similar to the first electrodes **210_8**, may end at the boundary of each subpixel PXn so as not to lie in another subpixel PXn. When some of the electrodes **210_8** and **220_8** are floating electrodes, the light-emitting elements **300** disposed between them may be partially connected in series in addition to the parallel connection. The outer bank **430** may be disposed at the boundary between the subpixels PXn neighboring each other in the first direction DR1 and may extend in the second direction DR2. The outer bank **430** may also be disposed at the boundary between the subpixels PXn neighboring each other in the second direction DR2 and may extend in the first direction DR1. The outer bank **430** is the same as that described above with reference to FIG. **3**. In addition, first contact electrodes **261_8** and second contact electrodes **262_8** included in the display device **10_8** shown in FIG. **34** are substantially the same as those included in the display device **10** shown in FIG. **3**.

(242) In FIG. **34**, two first electrodes **210_8** and two second electrodes **220_8** are disposed and are alternately spaced apart from each other. However, the present disclosure is not limited thereto, and some electrodes may be omitted, or more electrodes may be disposed in the display device **10_8**.

(243) In the display device **10**, the first electrode **210** and the second electrode **220** may not necessarily extend in one direction. The first electrode **210** and the second electrode **220** of the display device **10** are not limited to a particular shape as long as they are spaced apart from each other to provide a space in which the light-emitting elements **300** are disposed.

(244) FIG. **35** is a plan view of a pixel of a display device according to an embodiment.

(245) Referring to FIG. **35**, a first electrode **210_9** and a second electrode **220_9** of a display device **10_9** according to an embodiment may be at least partially curved, and a curved area of the first electrode **210_9** and a curved area of the second electrode **220_9** may be spaced apart to face each other. The display device **10_9** shown in FIG. **35** is different from the display device **10** shown in FIG. **2** in the shape of each of the first electrode **210_9** and the second electrode **220_9**. Therefore, any redundant description will be omitted, and differences therebetween will be primarily described below.

(246) The first electrode **210_9** of the display device **10_9** shown in FIG. **35** may include a plurality of holes (or openings) HOL. For example, as illustrated in the drawing, the first electrode **210_9** may include a first hole HOL1, a second hole HOL2, and a third hole HOL3 arranged along the second direction DR2. However, the present disclosure is not limited thereto, and the first electrode **210_9** may include a greater or fewer number of holes HOL or may include only one hole HOL. An embodiment in which the first electrode **210_9** includes the first hole HOL1, the second hole HOL2, and the third hole HOL3 will be described below as an example.

(247) In an embodiment, each of the first hole HOL1, the second hole HOL2, and the third hole HOL3 may have a circular planar shape. Accordingly, the first electrode **210_9** may include a curved area formed by each of the holes HOL and may face the second electrode **220_9** in the curved area. However, this is merely an example, and the present disclosure is not limited thereto. Each of the first hole HOL1, the second hole HOL2, and the third hole HOL3 is not limited to a

particular shape as long as it can provide a space in which the second electrode **220_9** is disposed, as will be described later, and may have various planar shapes, such as an elliptical shape and a quadrangular or more polygonal shape.

(248) The second electrode **220_9** may be provided in plural in each subpixel PXn. For example, three second electrodes **220_9** may be disposed in each subpixel PXn to correspond to the first through third holes HOL1 through HOL3 in the first electrode **210_9**. The second electrode **220_9** may be located in each of the first through third holes HOL1 through HOL3 and may be surrounded (e.g., surrounded along a periphery) by the first electrode **210_9**.

(249) In an embodiment, the holes HOL in the first electrode **210_9** may have curved outer surfaces, and the second electrodes **220_2** disposed in the holes HOL in the first electrode **210_9** may have curved outer surfaces and may be spaced apart from the first electrode **210_9** to face the first electrode **210_9**. As illustrated in FIG. 35, the first electrode **210_9** may include the holes HOL having a circular shape in a plan view, and the second electrodes **220_9** may have a circular shape in a plan view. The curved surfaces of the areas in which the holes HOL in the first electrode **210_9** are formed may be spaced apart from the curved outer surfaces of the second electrodes **220_9** to face them. For example, the first electrode **210_9** may surround the outer surfaces of the second electrodes **220_9**.

(250) As described above, light-emitting elements **300** may be disposed between the first electrode **210_9** and the second electrodes **220_9**. The display device **10_9** according to the current embodiment may include the second electrodes **220_9** having a circular shape and the first electrode **210_9** surrounding the second electrodes **220_9**, and the light-emitting elements **300** may be arranged along the outer surfaces of the second electrodes **220_2**. Because the light-emitting elements **300** extend in one direction as described above, the light-emitting elements **300** arranged along the curved outer surfaces of the second electrodes **220_9** in each subpixel PXn may be disposed such that their extending directions face different directions. Each subpixel PXn may have various light exit directions according to directions in which the extending directions of the light-emitting elements **300** face. In the display device **10_9** according to the current embodiment, because the first electrode **210_9** and the second electrodes **220_9** are disposed to have a curved shape, the light-emitting elements **300** disposed between them may face different directions, and lateral visibility of the display device **10_9** may be improved.

(251) In concluding the detailed description, those skilled in the art will appreciate that many variations and modifications can be made to the embodiments described herein without departing from the aspects and features of the present disclosure. Therefore, the embodiments of the present disclosure described herein are provided in a generic and descriptive sense and not for purposes of limitation.

Claims

1. A light-emitting element comprising: a first semiconductor layer and a second semiconductor layer; an active layer between the first semiconductor layer and the second semiconductor layer; and an electrode layer on the second semiconductor layer and having a first surface and a second surface facing the first surface, the second surface contacting the second semiconductor layer, wherein one of the first surface and the second surface of the electrode layer comprises a protrusion pattern protruding from at least a portion thereof and the other one of the first surface and the second surface of the electrode layer is flat, and wherein the protrusion pattern comprises: a first protrusion; a second protrusion spaced apart from the first protrusion; a third protrusion spaced apart from the first protrusion and the second protrusion and extending around a periphery of an outer surface of the second protrusion; a first depression between the first protrusion and the second protrusion; and a second depression between the second protrusion and the third protrusion.
2. The light-emitting element of claim 1, wherein the second protrusion extends around a periphery

of the first protrusion.

3. The light-emitting element of claim 1, wherein the first protrusion has an area having the same width as that of the second protrusion.

4. The light-emitting element of claim 3, wherein the first protrusion has an area having a width equal to a distance between the first protrusion and the second protrusion.

5. The light-emitting element of claim 1, wherein the protrusion pattern comprises a first protrusion pattern formed on the first surface, and wherein the light-emitting element extends in a direction and has a first end surface on which the first protrusion pattern is formed and a second end surface which is a surface of the first semiconductor layer.

6. The light-emitting element of claim 5, wherein the first end surface and the second end surface have different roughness.

7. The light-emitting element of claim 5, further comprising an insulating film which at least partially surrounds side surfaces of the first semiconductor layer, the second semiconductor layer, and the active layer.

8. A light-emitting element comprising: a first semiconductor layer and a second semiconductor layer; an active layer between the first semiconductor layer and the second semiconductor layer; and an electrode layer on the second semiconductor layer and having a first surface and a second surface facing the first surface, the second surface contacting the second semiconductor layer, wherein at least any one of the first surface and the second surface of the electrode layer comprises a protrusion pattern protruding from at least a portion thereof, wherein the electrode layer comprises a second protrusion pattern which is formed on the second surface to contact the second semiconductor layer, and wherein at least a portion of a surface of the second semiconductor layer that contacts the second surface of the electrode layer is depressed to contact protrusions of the second protrusion pattern.

9. A display device comprising: a substrate; a first electrode and a second electrode on a first surface of the substrate, the second electrode being spaced apart from the first electrode; and a light-emitting element between the first electrode and the second electrode and comprising a pattern part formed on an end surface, wherein the light-emitting element comprises: a first semiconductor layer and a second semiconductor layer; an active layer between the first semiconductor layer and the second semiconductor layer; and an electrode layer on the second semiconductor layer and comprising a protrusion pattern protruding from at least a portion of a surface in a direction parallel to the first surface of the substrate.

10. The display device of claim 9, wherein the protrusion pattern comprises a first protrusion, a second protrusion spaced apart from the first protrusion, and a depression between the first protrusion and the second protrusion.

11. The display device of claim 10, wherein the second protrusion extends around a periphery of the first protrusion.

12. The display device of claim 10, wherein the light-emitting element comprises a first light-emitting element comprising a first protrusion pattern and a second light-emitting element comprising a second protrusion pattern different from the first protrusion pattern.

13. The display device of claim 12, wherein the first protrusion pattern has a different number of protrusions than the second protrusion pattern.

14. The display device of claim 12, wherein a width of each protrusion in the first protrusion pattern is different from a width of each protrusion in the second protrusion pattern.

15. The display device of claim 9, wherein a first end of the light-emitting element is electrically connected to the first electrode, and a second end is electrically connected to the second electrode.

16. The display device of claim 15, wherein the electrode layer of the light-emitting element is electrically connected to the first electrode, and the first semiconductor layer is electrically connected to the second electrode.

17. A display device comprising: a first electrode and a second electrode spaced apart from the first

electrode; a light-emitting element between the first electrode and the second electrode and comprising a pattern part formed on an end surface, a first end of the light-emitting element being electrically connected to the first electrode, a second end of the light-emitting element being electrically connected to the second electrode, the light-emitting element comprising: a first semiconductor layer and a second semiconductor layer; an active layer between the first semiconductor layer and the second semiconductor layer; and an electrode layer on the second semiconductor layer and comprising a protrusion pattern protruding from at least a portion of a surface; a first contact electrode contacting the first end of the light-emitting element and the first electrode; and a second contact electrode contacting the second end of the light-emitting element and the second electrode.

18. The display device of claim 17, wherein the first contact electrode contacts the electrode layer of the light-emitting element, and the second contact electrode contacts the first semiconductor layer.
